

SID: 由自信号驱动的多大语言模型辩论

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摘要

大语言模型 (LLM) 已在多个应用领域展现出令人瞩目的能力。近年来，研究者提出多大语言模型智能体辩论 (Multi-LLM Agent Debate, MAD)，通过让多个模型反复讨论并迭代修正回答来提升性能。然而，现有 MAD 方法大多侧重于利用外部结构，例如辩论图或 LLM-as-a-Judge，而忽视了生成过程中天然产生的**自信号**，如 token logits 与注意力。这样的忽略会带来冗余计算，并可能造成性能下降。

本文将重点转向多 LLM 辩论中的自信号，提出**由自信号驱动的多 LLM 辩论框架** (Self-Signals Driven Multi-LLM Debate, SID)。该方法利用两类自信号：**模型级置信度**与**token 级语义焦点**，以自适应地引导辩论过程。具体而言，在模型层面，SID 允许高置信度智能体提前退出；在 token 层面，SID 基于注意力机制压缩辩论中的冗余内容。我们在多种 LLM 与多模态 LLM 上、针对多个具有挑战性的基准进行了评估。实验结果表明，SID 不仅在准确率上优于现有 MAD 技术，而且还能降低 token 消耗，说明利用自信号能够同时提升多智能体辩论系统的性能与效率。代码将公开于 <https://github.com/xuhang2019/SID>。

1 引言

大语言模型 (LLM) 已经在科学、技术、工程、数学 (STEM) 问题以及复杂推理任务等众多领域展示出强大的能力。多模态大语言模型 (MLLM) 的出现又将这种潜力扩展到了视觉输入场景。然而，当前模型依然存在固有限制，例如不准确和幻觉问题。

多智能体辩论 (MAD) 提供了一条与直接提升单模型能力相正交的路径：多个智能体先分别作答，再在后续回合中相互讨论、更新答案。尽管如此，MAD 也面临一个核心难题：辩论过程中常出现大量重复内容和反复确认的共识点。这不仅浪费计算资源，还会引入信息噪声，甚至影响最终判断。此外，随着基础模型能力不断增强，迭代讨论带来的 token 开销与实际收益之间的矛盾愈发明显，成为当代 MAD 研究中的中心问题。

为缓解这一问题，已有工作大体从两条路径进行优化：(1) **结构优化**，例如采用更复杂的提示策略、使用稀疏辩论图减少通信、或将智能体聚为局部辩论小组；(2) **历史管理**，例如对先前讨论做摘要，或让智能体自行生成置信分数。尽管这些方法能够从外部改善信息流动效率，但往往仍然依赖容易出错的外部机制，例如评审模型幻觉或摘要失真。由此我们提出一个自然的问题：能否不依赖这些容易出错的外部装置，而是直接利用智能体生成过程中更可靠的内部自信号，来阻止不必要且可能低效的辩论？

基于这一动机，本文提出一种在 LLM 推理过程中直接利用自信号的新框架。我们抽取两类互补信号：**模型级置信度**与**token 级语义焦点**，分别用于区分“需要继续辩论”的情况与“应当压缩保留”的关键内容，从而同时提升辩论质量与效率。模型级置信度来自模型对初始答案输出概率分布的估计，用于判断模型对当前回答的确信程度。我们据此设计了**早退机制**：当模型已足够自信时，不再启动后续辩论，以减少潜在噪声与冗余。token 级语义焦点则由面向“分歧”的提示条件下的注意力模式提取，用于识别辩论内容中与不同智能体分歧最相关的片段。我们提取并重构这些高注意力片段，形成更紧凑的上下文，由此得到一种新的**压缩机制**：既保留关键争议点，又显著降低 token 开销。

整合上述两种机制后，我们提出统一的 SID 框架，用模型自身的认知信号动态决定辩论是否应继续、以及应保留哪些内容。我们在 MMLUpro、Math、GPQA、ScienceQA 与 MMStar 等多个基准上

测试了该方法。SID 在绝大多数场景中都优于现有 MAD 方法，并且在部分设置下实现了最高约 40% 的 token 节省。这表明，在多智能体系统中利用内部信念信号，能够同时优化性能与效率。

本文贡献如下：

- 提出一种利用 LLM 生成过程自信信号增强智能体辩论的新思路，摆脱了对容易出错的外部机制的过度依赖。
- 提取两类关键自信信号：模型级置信度与 token 级语义焦点，并分别设计早退机制与压缩机制，以减少冗余、提升辩论质量。
- 将两种机制整合为统一而高效的 SID 框架，并在多个 LLM 与 MLLM 基准上验证了其显著优势。

2 相关工作

2.1 推理增强

为提升 LLM 的推理能力，研究者提出了多种方法。早期工作主要通过 Chain-of-Thought (CoT) 提示引导模型进行逐步推理，或通过生成多条推理路径、再以自一致性 (self-consistency) 或多轮自反思来选择更优答案。随后研究发现，模型的自我纠错能力虽然重要，但仍然有限，容易导致推理质量停滞。这在一定程度上推动了多智能体协作范式，尤其是多智能体辩论 (MAD) 的发展：通过引入外部视角与动态反馈，弥补单模型自反思的局限。与上述工作不同，本文关注的不是单纯用提示改善推理，而是利用模型自身的内部信号，在 token 层面对上下文进行优化，从而同时改善性能与 token 效率。

2.2 不确定性分析

LLM 的不确定性通常被划分为随机性不确定性 (aleatoric) 与认知性不确定性 (epistemic)。考虑到问答、数学、科学等任务本身结构明确，近期研究更多聚焦于对认知性不确定性的量化。主流方法包括：(1) 基于概率的指标，如 token 级熵与负对数似然；(2) 基于集成的方法，如 Monte Carlo 采样与贝叶斯方法；(3) 基于语言化表达的方法，即提示模型自行报告置信度。其中，概率类方法尤其有吸引力，因为它们能够无缝嵌入生成过程，无需额外多次生成，因此不会引入显著的 token 开销。近来的工作也开始在交互式场景中研究智能体级不确定性，强调其在学习与输出控制中的作用。本文的方法与这一方向一致，将自信信号视为动态的智能体级不确定性指示器，用于控制智能体是否参与辩论。

2.3 多 LLM 辩论系统

早期多 LLM 辩论常采用角色扮演式设置，这在协作任务中已显示出优势，但在某些问题求解场景下并不总是最合适。MAD 通过引入外部视角来丰富系统推理能力；DMAD 利用专门设计的提示策略增强智能体行为差异；S²-MAD 引入选择性稀疏机制，让智能体依据内部线索决定是否参与；CortexDebate 则让智能体兼任自评审角色、输出置信分数，并构造动态稀疏辩论图。这些方法主要通过优化外部状态来提升性能，例如优化通信结构或依赖 LLM-as-a-Judge。相比之下，本文提供了一个可与既有方法互补的新角度：不是主要优化通信拓扑，而是将模型内部自信信号直接融入辩论过程。

3 预备知识

本节先介绍朴素的多智能体辩论范式。设 V 为词表，Tok 为分词器。给定查询 (可为自然语言、图像与文本组合) Q ，记 $x = \text{Tok}(Q)$ 为分词后的提示。自回归 LLM M 生成响应序列 $\mathbf{y} = (y_1, \dots, y_m)$ ，对应每一步的 logits 为 $\ell_t \in \mathbb{R}^{|V|}$ ，概率分布为 $\pi_t = \text{softmax}(\ell_t)$ 。一个辩论包含 n 个智能体 $A = \{1, \dots, n\}$ ，并在 $t = 0, 1, \dots, T$ 的多个回合中进行。记 $y_t^{(j)}$ 为第 j 个智能体在第 t 轮的回答；第 1 轮是没有辩论

上下文的初始作答。对智能体 j 而言，第 $t+1$ 轮的输入由原始查询、自身上一轮回答以及其他智能体上一轮回答拼接而成：

$$x_{t+1}^{(j)} = \text{Tok}(Q \parallel y_t^{(j)} \parallel (\text{Concat}_{k \neq j} y_t^{(k)})). \quad (1)$$

其中， $\parallel$ 与 Concat 都表示提示片段之间的拼接。

4 方法

上述朴素框架面临两个突出问题：冗余严重、效率偏低。为此，本文提出 SID (Self-Signal Driven Debate) 框架：直接利用推理过程中可获得的内部置信信号，自适应地引导多 LLM 辩论。具体来说，SID 使用两类自信信号：**模型级置信度**与 **token 级语义焦点**。前者来自初始回答输出 logits 所反映的置信程度，用于决定是否早退；后者来自在“分歧导向”提示下形成的自注意力图，用于定位高价值内容并进行压缩。图 1 与算法 1 概述了整个流程。

Algorithm 1 Self-Signal Driven Debate (SID)

Require: Query Q ; LLM agents $\{M_j\}_{j=1}^m$; rounds N ; confidence threshold θ ; top- p ratio ρ ;

- 1: $Y_1 \leftarrow M_1(Q)$
- 2: $u \leftarrow \phi_U(U(Y_1))$ ▷ Model-level Confidence(Sec. 4.1)
- 3: **if** $u \leq \theta$ **then**
- 4: **return** Y_1
- 5: **end if** ▷ early exit
- 6: $Y_1 \leftarrow \{y_1\}$; $y_1^{(j)} \leftarrow M_j(Q)$ for $j=2..m$
- 7: **for** $t = 2$ **to** N **do**
- 8: **for** $j = 1$ **to** m **do**
- 9: $X_t^{(j)} \leftarrow (Q \parallel y_{t-1}^{(j)} \parallel [\text{PROMPT}] \parallel \text{Concat}_{k \neq j}(Y_{t-1}^k))$
- 10: $A_t^{(j)} \leftarrow \text{FORWARDATTENTION}(M_j, X_t^{(j)})$ ▷ forward only
- 11: $\tilde{C}_t^{(j)} \leftarrow \text{TopP}(A_t^{(j)}, \rho)$
- 12: $S_t^{(j)} \leftarrow \text{SemanticPreserve}(\tilde{C}_t^{(j)})$ ▷ Sec. 4.2
- 13: $y_t^{(j)} \leftarrow M_j(Q \parallel y_{t-1}^{(j)} \parallel S_t^{(j)})$ ▷ generate with compressed context
- 14: **end for**
- 15: $Y_t \leftarrow \{y_t^{(j)}\}_{j=1}^m$
- 16: **end for**
- 17: **return** Y_N

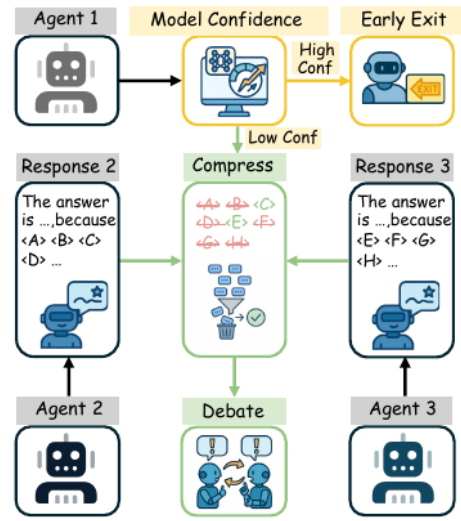


Figure 1: Overall framework of SID

图 1: 图 1 (保留原图): SID 的总体框架与算法 1。

4.1 基于模型级置信度的早退机制

我们首先引入一个早退机制，以减少冗余辩论。其核心直觉是：当单个模型本身已经对回答具有很高把握时，跨模型讨论往往没有必要。关键在于如何提取有效的置信分数。直观上，输出分布越尖锐（也即熵越低），模型就越自信。基于这一动机，并遵循既有做法，本文采用两种 token 级不确定性指标来估计生成答案 y 的置信度：

- 熵: $\text{Ent}(\pi_t) = - \sum_{v \in V} \pi_{t,v} \log \pi_{t,v}$;
- 负对数似然: $\text{NLL}_t = - \log \pi_{t,y_t}$ 。

为了将 token 级指标聚合为序列级置信度，作者考察了四种汇聚策略：对所有 token 求平均 (average)、取最大值 (max)、使用首 token 的值 (first)，以及使用倒数第二个 token 的值 (penultimate)。这样便得到 8 个置信度特征（熵 4 个、NLL 4 个）。不同特征关注不确定性的不同侧面，例如 max 更强调“最坏情况”歧义，而 penultimate 更贴近自回归生成最后阶段的推理稳定性。将这 8 个量拼接后即可形成向量 $U(y)$ ，实验表明它对于区分正确与错误答案具有统计显著性。

获得置信度后，下一步是如何把它用于多 LLM 辩论中的二元决策边界。本文提出两种可直接替换的策略：

词表自适应阈值 最直接的做法是为所有模型统一设定一个固定阈值，但这种方法效果并不理想。原因在于熵与 NLL 的数值大小会受到模型词表大小 $|V|$ 的天然影响：在均匀分布假设下，二者都等于 $\log |V|$ 。因此，词表越大，这些值自然越高，如果所有模型都用同一阈值，就会产生不公平与不稳定。基于此，文中提出词表自适应阈值：

$$\theta(V) = \alpha \log |V|, \quad \text{当且仅当 } \phi_U(U(y)) \leq \theta(V) \text{ 时终止。} \quad (2)$$

其中， $\alpha > 0$ 为超参数， ϕ_U 是用于过滤噪声置信度特征的算子。该策略使不同词表规模的模型能够在更加公平、一致的标准下进行置信评估。

校准置信度 虽然词表自适应阈值简单而稳健，但它依赖于“分布近似均匀”的假设。为捕捉更细腻的置信信息，作者又提出一种替代方案：在小型留出集上训练一个轻量级非线性分类器 $C: \mathbb{R}^d \rightarrow [0, 1]$ ，输入为置信度向量，输出为与正确性对齐后的标量置信度：

$$\text{当且仅当 } C(U(y)) \geq \tau_c, \quad \tau_c \in (0, 1) \text{ 时终止。} \quad (3)$$

实践中的选择 在实际实验中，作者最终选择了**词表自适应阈值**作为 gating 机制，因为它无需训练、实现简单，同时性能已经足够强。具体做法是：在首轮回答之后，如果某个智能体的置信度足够高，则立即提前退出，意味着系统已经对当前答案有较高把握；反之，如果模型置信度较低，则说明问题具有较高难度，更有必要启动多智能体辩论。

4.2 基于 token 级语义焦点的自适应压缩

除了模型级置信度之外，SID 还利用另一类自信号来提升辩论效率，即 **token 级语义焦点**。随着辩论推进，多智能体累积起来的上下文常会越来越长，其中包含大量重复与冗余内容。这些冗余会降低信噪比，既影响效果，也拖慢推理。一种常见做法是使用 LLM-as-a-Judge 对历史内容做摘要，但这种办法受基础模型摘要能力限制，容易出现幻觉或信息遗漏。为此，本文转而使用 Transformer 模型内部固有的注意力机制，把它作为内部信号来实现 token 级压缩。

提示条件下的注意力提取 给定查询 Q 、智能体 j 在上一轮的回答 $y_t^{(j)}$ ，以及其他智能体的回答 $\{y_t^{(k)} : k \neq j\}$ ，拼接输入写为：

$$x_{t+1}^{(j)} = \text{Tok}\left(Q \parallel y_t^{(j)} \parallel [\text{PROMPT}] \parallel (\text{Concat}_{k \neq j} y_t^{(k)})\right). \quad (4)$$

其中，[PROMPT] 是任务指令。文中使用的提示为：

“找出它们与你自身推理不一致的关键点。聚焦这些分歧，并判断哪条推理路线更优。”

该提示将模型注意力显式引向辩论中存在语义冲突的片段。设 Q 为注入提示部分对应的 token 位置集合， C 为其他智能体回答对应的位置集合， $A^{(l,h)} \in [0, 1]^{L \times L}$ 为第 l 层第 h 个头的注意力得分。对于 $c \in C$ ，其提示条件下的语义焦点分数定义为：

$$s(c) = \max_{l,h} \max_{q \in Q} A_{q,c}^{(l,h)}. \quad (5)$$

它表示：在所有层、所有头中，提示 token 指向位置 c 的最大注意力权重，也即该 token 与“分歧判断”任务的相关程度。

带语义保持的压缩 仅凭 token 级分数 $s(c)$ 可以细粒度识别关键内容，但若直接选取若干高分 token，容易得到被切碎的短语甚至残缺句子，从而破坏模型对压缩后输入的理解。为此，作者加入了**语义保持**启发式：将高注意力 token 扩展到完整的句子或从句层级。首先选取上下文中 top- p 比例的 token，

形成 $\hat{C} = \text{Top}_p\{(c, s(c))\}_{c \in C}$ ；然后利用分词器偏移映射 $\Psi : c \mapsto [T_a(c), T_b(c)]$ 将这些 token 映射回文本区间，合并重叠片段，并向句界扩展，以保留语义完整性。作者将这一过程记为 `SemanticPreserve`：

$$S = \text{SemanticPreserve}\left(\text{Merge}\left(\{\Psi(c) : c \in \hat{C}\}\right)\right). \quad (6)$$

于是，与这些片段对应的压缩文本记为 $\text{Text}(S)$ ，下一轮输入可写成：

$$\hat{x}_{t+1}^{(j)} = \text{Tok}\left(Q \parallel y_t^{(j)} \parallel \text{Text}(S)\right). \quad (7)$$

用 $\text{Text}(S)$ 替代完整历史后，系统可以在保留争议焦点的同时显著压缩 token 数。

4.3 整体框架

综合以上两类机制，SID 的流程如下：初始生成后，每个智能体先根据 logits 计算自身的模型级置信度；若足够自信，则提前退出辩论；若不够自信，则进入辩论流程。此时，系统用“分歧导向”的提示引导注意力，定位其他智能体回答中最具语义价值的片段，并利用语义保持策略将其重构为简洁上下文供下一轮使用。借助生成时刻的不确定性与注意力驱动的语义聚焦，SID 能够按照模型自己的内部信念状态自适应地决定辩论轨迹，在无需额外训练的前提下兼顾高效率与稳健性。

5 实验

5.1 实验设置

任务与基准 作者同时报告了 LLM 与 MLLM 两类任务结果。对 LLM，选用 MMLUpro 与 Math 数据集，它们覆盖了不同领域、不同难度的问题求解任务；对 MLLM，则选用 ScienceQA 与 MMStar。与既有工作一致，每个数据集随机采样 100 个问题进行评估。对于 ScienceQA，按照先前工作做法使用 lecture 与 hint 作为额外文本信息；其余数据集默认采用 zero-shot 提示设置。

模型 为覆盖不同类型的基础模型，实验同时考察了通用模型与偏推理模型。LLM 方面，使用 LLaMA-3.1-Instruct-8B (LLaMA3.1-8B) 与较新的 GPT-OSS-20B；MLLM 方面，使用 LLaVA-v1.6-Vicuna-13B (文中记为 LLaVA1.6-13B) 以及 GLM4.1V-Thinking (GLM4.1V)。

实现细节 为公平比较，作者遵循先前工作设置，在 SID、MAD 与 DMAD 中统一采用 $n = 3$ 个智能体、 $N = 2$ 轮辩论；self-consistency 的采样数设为 3。此外，还结合了 step-back prompting 与 self-contrast 等推理增强方式，与 IO（直接输出）和 COT 一并比较。对于模型级置信度，NLL-max 阈值中的 α 在不同模型上取不同值：推理型模型为 1.0，通用模型为 0.5，MLLM 为 0.25。为减轻 attention sink 与特殊 token 对 logits 的影响，文中将 $\phi(U)$ 设为 NLL 与熵中的最大值，并在计算模型级置信度时排除某些位置特征。校准置信度方法则在 50 个留出样本上训练， $\tau_c = 0.9$ 。

评测指标 Math 数据集采用官方 exact match；其余多项选择式问答数据集均以准确率作为评价指标。

5.2 主要结果

Table 1: Performance comparison across different LLMs for various datasets (Math subsets and MMLUpro). SID-v and SID-c denote our method using the vocabulary-adaptive threshold and calibrated confidence, respectively, to implement the early-exit mechanism. (see Sec 4.1 for details)

Model	Method	Alg.	C&P	Geo.	Int.A.	Num	Pre.A	Pre.C.	MMLUpro	Avg
LLaMA3.1-8B	COT	61	38	34	14	37	54	28	39	38.13
	IO	65	37	35	15	46	59	28	25	38.75
	SBP Zheng et al. 2024	46	28	21	12	33	46	24	15	28.13
	Self-Consistency Wang et al. 2023	58	25	32	12	40	55	25	45	36.50
	Self-Contrast Zhang et al. 2024	54	36	27	11	31	53	27	36	34.38
	MAD Du et al. 2024	61	36	36	16	37	60	29	41	39.50
	DMAD Liu et al. 2024c	55	36	32	13	36	58	26	39	36.88
	SID-v	67	43	40	18	41	64	31	47	43.88
GPT-OSS-20B	SID-c	67	43	39	20	41	65	30	47	44.00
	COT	85	81	56	36	70	84	44	61	64.63
	IO	85	81	60	40	74	87	42	64	66.63
	SBP Zheng et al. 2024	65	64	44	37	16	73	11	26	42.00
	Self-Consistency Wang et al. 2023	75	67	44	31	70	79	23	69	57.25
	Self-Contrast Zhang et al. 2024	84	75	65	36	67	88	35	65	64.38
	DMAD Liu et al. 2024c	91	90	73	51	66	89	47	65	71.50

图 2: 表 1 (保留原表): 不同 LLM 在多个数据集上的性能比较。SID-v 使用词表自适应阈值, SID-c 使用校准置信度。

整体性能 表 1 与表 2 (保留原图) 展示了不同模型与数据集上的总体结果。SID 在大多数场景中都取得了最佳性能, 表现出较强的有效性。与先前结论一致, MAD 类方法通常优于 self-consistency 等单模型推理增强基线。另一个值得注意的现象是, 采用词表自适应阈值 (SID-v) 与采用校准置信度 (SID-c) 时, 性能非常接近, 说明简单阈值已经能够较好逼近学习到的决策边界。考虑到其无需训练且实践效果优良, 作者建议在真实应用中优先使用 SID-v。

Table 2: Performance on Sci.QA and MMStar based on MLLMs LLaVA1.6-13B and GLM4.1V.

Model	Method	Sci.QA	MMStar
LLaVA1.6-13B	CoT	63	11
	IO	62	9
	Self-Consis	63	11
	MAD	65	12
	SID-v	65	14
GLM4.1V	CoT	83	29
	IO	83	32
	Self-Consis	84	29
GLM4.1V	MAD	90	47
	SID-v	91	54
	SID-c	91	54

Table 3: Ablation Study of SID on MMLUpro based on LLaMA3.1-8B.

Method	Accuracy	Token Ratio
Baseline Single-round CoT	37.67	0.17
Baseline MAD	39.50	1.00
Baseline MAD + Compression	41.67	0.73
Baseline MAD + Compression + Early Exit	46.83	0.53
SID w/o Semantic Preservation	34.50	0.46
SID w/o Early Exit w/ Token-level Summary	39.50	0.68
SID	46.83	0.53

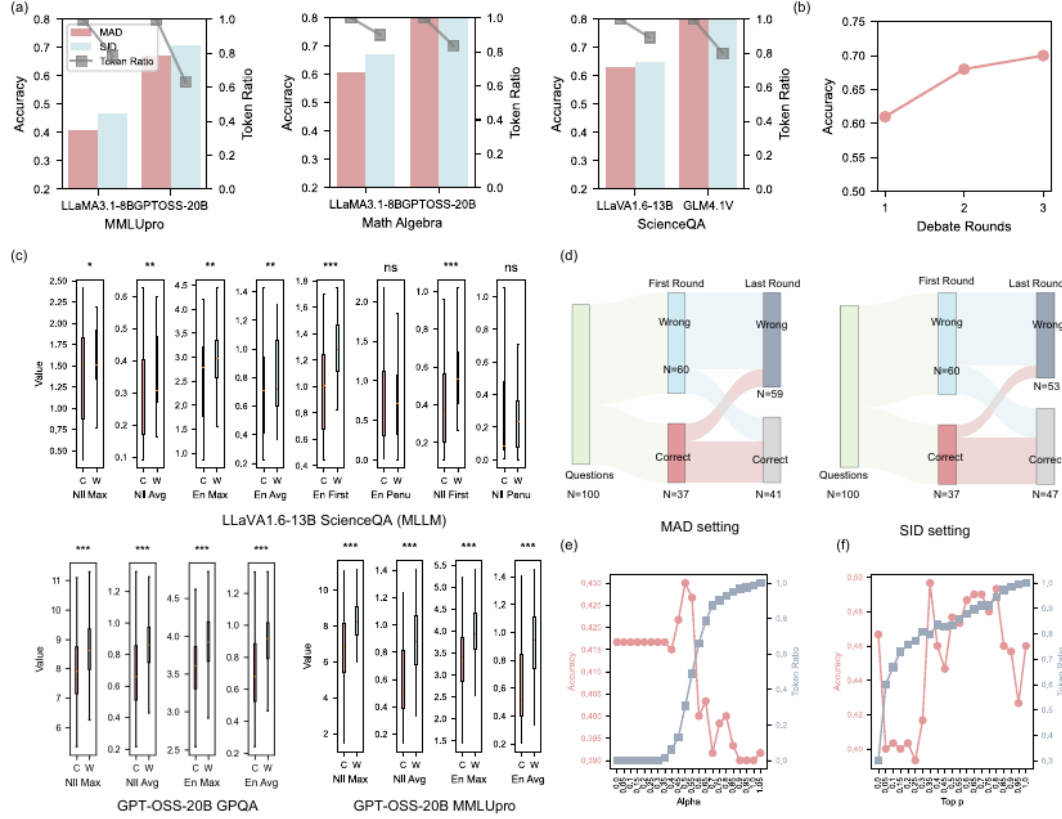


Figure 2: (a) Accuracy and token ratio comparison across strategies in MAD vs SID. (b) Performance with more debate rounds in LLM and MLLM. (c) Significance tests on model-level confidence.

图 3: 图 2 与表 2-3 (保留原图): 准确率、token 开销、统计显著性、错误修正流, 以及若干消融结果。

准确率与效率 图 2(a) 将 SID 与基线 MAD 在准确率和 token 消耗比两个维度进行比较, 其中 token ratio 以 MAD 设为 1 作为基准。结果显示, SID 在科学与推理类数据集上可实现最高约 40% 的 token 减少, 同时准确率更高, 说明其在效率与效果上都具有明显优势。文中还指出, 在 GPT-OSS 与 GLM4.1V 这类 thinking 模型上, token 压缩收益相对更难获得, 因此进一步的降耗更具价值。附录中的运行时间比较也显示, SID 推理时间显著更短, 进一步证明了其效率优势。图 2(b) 则展示了随着辩论回合增加, SID 仍能持续获得提升, 说明其具备良好的可扩展性。

统计显著性分析 图 2(c) 以及附录图 7-16 展示了模型级置信度特征的统计显著性。图中 C 与 W 分别表示回答正确与回答错误的样本组。无论是在 GPQA 还是 MMLUpro 上, 对于同一模型, SID 的置信阈值都在正确样本组中表现出较高稳定性 (例如 NLL max 约为 7.5), 说明模型级置信度信号具有较强的稳健性。

5.3 消融与分析

关键组件消融 作者基于 LLaMA3.1-8B 与 MMLUpro 对框架关键设计做了全面消融。结果表明，基线 MAD 的性能并不理想；在其基础上加入模型级置信度早退（第 4.1 节）和 token 级语义焦点压缩（第 4.2 节）后，效果与 token 效率都得到显著提升。进一步地，去掉语义保持组件会造成性能明显下降，说明仅保留稀疏 token 而不恢复语义完整性是不够的。作者还将 token 级压缩替换为自我摘要策略（self-summary），将其作为 LLM-as-a-Judge 范式的代表进行比较，结果略逊于 SID，表明直接利用生成过程中的自信信号更加高效且稳健。

词表自适应阈值 α 的消融 文中还考察了阈值超参数 α 与早退比例之间的关系。若 α 很小，则几乎所有问题都会被视作“不自信”，退出比例接近 0，此时系统退化为传统 MAD；若 α 很大，则几乎所有问题都会被视作“自信”，系统仅保留第一轮回答，退化为单模型设置。实验表明，对 LLaMA3.1-8B 而言， $\alpha = 0.5$ 是较优选择。

语义保持比例 p 的消融 在语义保持框架中，系统会选取上下文中 top- p 比例的 token 做进一步处理。实验结果显示， $p \approx 0.35$ 或 0.4 时表现最好。一个有趣现象是：当 $p > 0$ 但非常小时，性能可能反而不如完全不加入额外上下文；而当 p 过大时，由于保留了更多冗余一致信息，性能同样会下降。这说明，无论是信息不完整还是信息过度冗余，都会损害多 LLM 辩论的有效性。

5.4 可视化结果

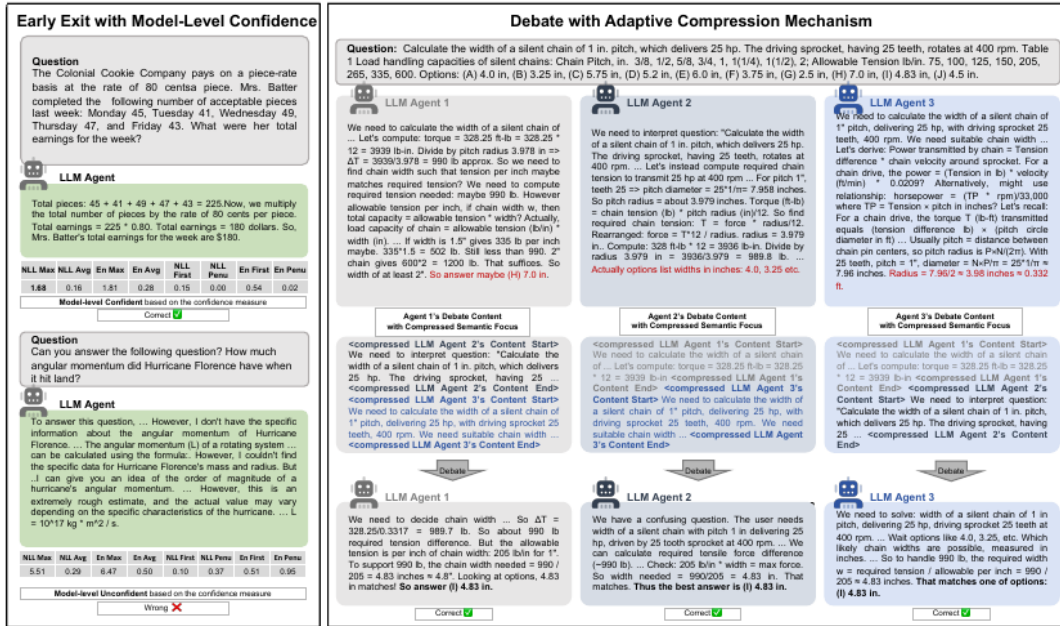


Figure 3: Case study of SID's debate process. (Left) On MMLUpro, SID exits early for a simple

图 4: 图 3 (保留原图): SID 辩论流程案例。左侧展示模型级置信度驱动的早退; 右侧展示 token 级语义焦点驱动的自适应压缩辩论。

为更直观地说明框架机制，作者在图 3 中展示了 SID 的可视化案例。左侧分支展示了早退机制：在一个简单的经济学算术题上，模型生成答案后被评估为高置信度（例如 $NLL\ Max = 1.68$ ），因此直接提前退出，且预测正确；而面对一个更复杂的物理问题时，模型被判定为低置信度（例如 $NLL\ Max = 5.51$ ），从而触发进一步辩论。右侧分支展示了自适应压缩机制：面对一题较难的物理问题，三个智能体在初始阶段都给出了错误回答；但在 token 级语义焦点驱动的压缩辩论中，它们逐步修正推理，

最终收敛到正确答案。文中还指出，与 MAD 相比，SID 显著减少了“原本正确却在辩论后变错”的情形，同时增加了“原本错误但经辩论后纠正”的案例，进一步说明了其有效性。

6 结论

本文提出 SID——一种利用 LLM 生成过程自信号来同时提升性能与效率的多 LLM 辩论框架。SID 集成了两类内部信号：一是模型级置信度，用于让高置信度智能体提前退出；二是 token 级语义焦点，用于基于注意力保留关键分歧点并压缩辩论历史。在多个 LLM 与 MLLM 基准上的实验表明，SID 兼具高性能与高效率，显示出将模型内部状态作为有效信号来引导协作式问题求解的巨大潜力，也为多智能体系统中的新范式提供了值得继续探索的方向。

可复现性声明

作者为保证结果可复现性做了较充分的说明。主文中的第 4 节、算法 1 与第 5.1 节给出了框架方法、对比基线、基准数据集、模型配置及评测设置；附录 C 中进一步补充了实现细节与完整算法。为帮助读者忠实复现该方法，论文还在表 4、表 5 与图 4 中给出了关键提示词与指令格式。作者认为，这些材料足以支持对本文研究的复现实验。

说明

以下页面保留原文：**参考文献与附录**。

Preprint. Under review.

REPRODUCIBILITY STATEMENT

Significant efforts have been made to ensure the reproducibility of our results. The implementation details of our framework are described in the main manuscript (Section 4, Algorithm 1, and 5.1), including methods, baselines, benchmarks, model configurations, and evaluation settings. Additional implementation details and the full algorithm are provided in Appendix C. To facilitate faithful replication of our method, we include detailed descriptions of the key prompts and instruction formats in Table 4, Table 5, and Figure 4. We believe these materials are sufficient to enable reproducibility of our study.

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Preprint. Under review.

A USE OF LARGE LANGUAGE MODELS (LLMs)

Large Language Models (LLMs) were used solely for language refinement and proofreading purposes. They were not involved in research ideation and methodology design. All scientific contributions and conceptual developments were carried out entirely by the authors. The LLM did not play a substantive role in shaping the research content and should not be considered a contributor.

B LIMITATION

Our method relies on internal model signals such as logits and attention maps, which limit direct applicability to public closed-source APIs. However, it remains well-suited for internal deployments of proprietary models, especially in multi-agent systems, and can serve as an intermediate reasoning layer prior to externalized API serving. Notably, many modern systems (e.g., GPT-5) already adopt multi-agent or tool-augmented architectures, making our approach broadly applicable and increasingly relevant.

C ALGORITHM AND IMPLEMENTATION

C.1 SEMANTIC PRESERVATION

The semantic preservation module plays a key role in restoring the semantic cohesion from the top- p selected sparse tokens based on a model’s self-signals (*i.e.*, attention mechanism). Specifically, the method selects the most relevant textual spans based on attention distribution, but ensures that these selections are semantically coherent when mapped back to natural language. The algorithm below (Algorithm 2) shows the main pipeline, and the example (Figure 6) illustrates the comparison between without and with semantic preservation.

Algorithm 2 Semantic-Preserving Compression

Require: Prompt text x with marked spans FOCUS, DISCUSSION; offset map $\mathcal{O}$; Top- p selected attention score $\mathcal{C}$

Ensure: Compressed prompt x'

- 1: $\mathcal{U} \leftarrow \text{EXTRACTUNITS}(x, \text{DISCUSSION})$ $\triangleright$ Sentence/clause-level segments
 - 2: $T \leftarrow \text{TOKENIZER}(x)$
 - 3: $\mathcal{S} \leftarrow \text{MAPTOKENSTOUNITS}(\mathcal{C}, \mathcal{U}, \mathcal{O})$
 - 4: $x' \leftarrow \text{REPLACESPAN}(x, \text{DISCUSSION}, \mathcal{S})$
 - 5: **return** x'
-

We begin by extracting semantically coherent units (e.g., sentences or clauses) from the DISCUSSION span using lightweight parsing heuristics, including punctuation or newline segmentation. This yields a set of candidate text fragments $\mathcal{U}$.

We then use the Top- p selected attention score $\mathcal{C}$ (from Algorithm 1) to select the top- p most relevant tokens from T . To preserve semantic interpretability, we map these selected tokens back to their enclosing segments in $\mathcal{U}$ using the token-to-text offset map $\mathcal{O}$. The resulting set of informative fragments $\mathcal{S}$ is used to replace the original DISCUSSION span, yielding a compressed prompt x' that retains critical disagreement signals while discarding redundant or low-relevance content.

In the multi-modal setting (e.g., MLLMs), token offsets may shift due to image-text fusion. We mitigate this by anchoring to stable textual markers in the FOCUS span to adjust $\mathcal{O}$ and maintain alignment.

This compression module is integrated into the overall SID framework to support efficient and interpretable multi-agent reasoning under token or latency constraints

Preprint. Under review.

C.2 PROMPT TEMPLATE

In multi-task evaluation settings, especially those involving factual or multiple-choice benchmarks, we observe that models frequently generate semantically correct answers but fail to conform to the expected output format. This discrepancy is particularly pronounced in open-ended LLMs, where prior supervised fine-tuning (SFT) phases may introduce implicit formatting preferences (e.g., `\boxed` in math domains).

To mitigate this, we prepend a task-specific *system prompt* that explicitly enforces the desired answer format. Our full prompting format is:

`<system prompt> + <question content> + <output instruction>`

This method proves especially helpful for models with weaker instruction-following capabilities (e.g., LLaMA3.1-8B) and significantly reduces post-hoc answer parsing failures. Another example is the GLM4.1-V thinking model. The default multiple choice response uses a special boxed token, such as `<|begin_of_box|>B<|end_of_box|>`. By emphasizing the answer returning with brackets in the system prompt, GLM4.1V thinking yields `<|begin_of_box|>(B)<|end_of_box|>`. This allows us to extract the result using brackets in a unified way. Table 4 lists the dataset-specific system prompts and the enforced answer formats used in our experiments.

Table 4: Dataset-specific system prompts and enforced output formats for answer extraction.

Dataset	System Prompt (Instruction)	Expected Output Format (for answer parsing)
MMLUpro	You are a trivia expert who knows everything. You are tasked to answer the following question. Give your final answer in the format of (X), e.g., (A).	(A), (B), etc.
Math	You are a math expert. You are tasked to determine the answer to the following question. Give your final answer in the form of <code>\boxed{answer}</code> in the last sentence of your response, e.g., <code>\boxed{[1, 3]}</code> .	<code>\boxed{...}</code>
GPQA	You are an expert in graduate-level science and mathematics. You will be presented with challenging questions designed to test your reasoning abilities. Your last sentence should be "The correct answer is (insert answer here)."	"The correct answer is (A)."
ScienceQA	You are a trivia expert who knows everything. You are tasked to answer the following question. Give your final answer in the format of (X), e.g., (A).	(A), (B), etc.
MMStar	You are an expert in multimodal task understanding, and your task is to answer the following questions. Give your final answer in the format of (X), e.g., (A)	(A), (B), etc.

Table 5: Dataset-specific output instruction prompts.

Dataset	Output Instruction
MMLUpro	Give your final answer in the format of '(X)'
Math	Give your final answer in the form of <code>\\boxed{answer}</code> at the end of your response, e.g., <code>\\boxed{[1, 3]}</code> .
GPQA	Your last sentence should be 'The correct answer is (insert answer here).' e.g., The correct answer is (A).
ScienceQA	Give your final answer in the format of '(X)'. You should only give one answer. For example, the answer is (A).
MMStar	Give your final answer in the format of '(X)'. You should only give one answer.



Figure 4: Details of reasoning augmentation prompt.

In terms of question content, we strictly follow the previous work Du et al. (2024); Liu et al. (2024c) in parsing the question to the chat template.

Moreover, we list the reasoning augmentation prompt (Figure 4 used in our experiments. Notably, Output Instructions should still be used after those prompts to enhance the ability to follow instructions.

Preprint. Under review.

D MORE RESULTS

In this section, we list many visualization results to illustrate the effectiveness of our SID methods.

Figure 6 Example comparison between w/o semantic preservation (red, brute-force token selection) and w/ semantic preservation (green, semantically coherent expansion). It can be observed that token-level semantic focus deletes irrelevant points (for example, point 2 in the solution was deleted), and our semantic preservation retains the semantic cohesion from the selected tokens.

A series of model-level confidence examples below can demonstrate the stable early exit threshold in the same model, and the statistical significance between the correct and wrong groups. Moreover, we also provide the correction flow from the first round to the last round.

Figure 7 Top: Model-level Confidence result on the Math Algebra dataset with LLaMA3.1-8B. Bottom: Correction flow with 8 deltas of confidence metrics. Unc: the answer remains unchanged between the first round and the last round, C2W: correct to wrong, W2C: wrong to correct.

Figure 8 Model-level Confidence result on the Math Counting and Probability dataset with LLaMA3.1-8B.

Figure 9 Model-level Confidence result on the Math Geometry dataset with LLaMA3.1-8B.

Figure 10 Model-level Confidence result on the Math Intermediate Algebra dataset with LLaMA3.1-8B.

Figure 11 Model-level Confidence result on the Math Number Theory dataset with LLaMA3.1-8B.

Figure 12 Model-level Confidence result on the Math Prealgebra dataset with LLaMA3.1-8B.

Figure 13 Model-level Confidence result on the Math Precalculus dataset with LLaMA3.1-8B.

Figure 14 Model-level Confidence result on the MMStar dataset with GLM4.1V.

Figure 15 Model-level Confidence result on the MMLUpro dataset with GPT-OSS-20B. Bottom: Correction flow with 8 deltas of confidence metrics.

In addition, a number of model-level early exit cases are provided here to show the confident and overconfident cases. It can be observed that the model partially analyzes the problem in overconfident cases.

Figure 17 Examples of model-level early exit cases in the MMLUpro dataset.

Figure 18 Examples of model-level early exit cases in the ScienceQA dataset.

Figure 19 Examples of model-level early exit cases in the Math dataset.

Figure 20 Examples of model-level early exit cases in the GPQA dataset.

Subsequently, Figure 21 and Figure 22 display that the token-level semantic focus module compresses the contents and assists agents in correcting their answers.

Preprint. Under review.

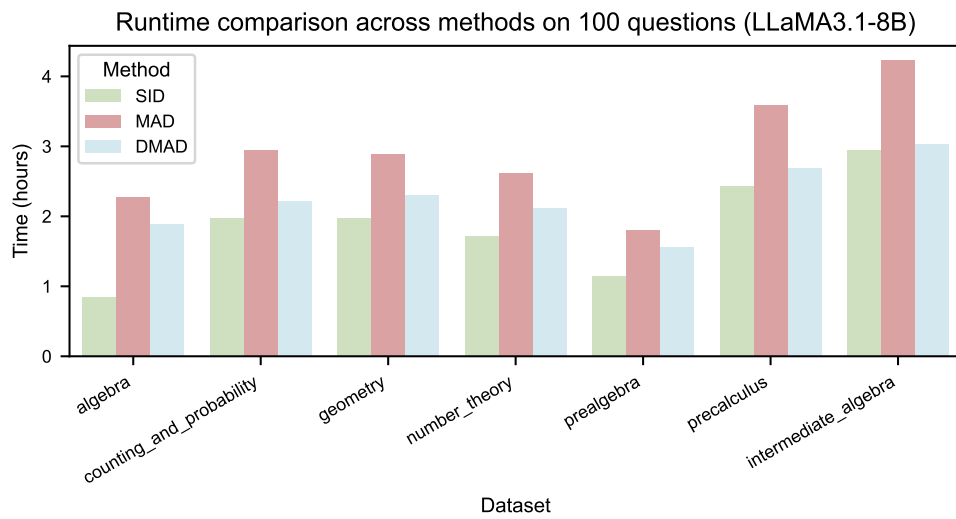


Figure 5: Running time comparison on math datasets with LLaMA3.1-8B, running on single A100 80GB GPU

Examples of Semantic Preservation

<Selected Text: w/o Semantic Preservation>

One agent solution:

```To solve this, let's break it down by step:

1. Both parents have the same meaning they are both they can only that also zy they can only).

3. Since are but not).

4 the probability that each of children will bezy islor %), because they will only from both parents.

So, the correct answer is (F) 1.

```

One agent solution:

```To solve this let's consider the possibleotypes andotypes of the and both parents are we can their the a

2 Theotypes of )

3 parents the possible can

4 The probability of each child is the of which is because each has of probability of2).

So, the probability that each children will be affected equals 4 or 25%. Therefore, the answer is (I).

```

<Selected Text: w/ Semantic Preservation>

One agent solution:

```To solve this, let's break it down step by step:

1. Both parents have the same autosomal recessive disorder, meaning they are both homozygous recessive (rr) for the disorder. When two homozygous recessive parents (rr) have children, they can only produce offspring that are also homozygous recessive (rr) because they can only contribute their recessive alleles (r).

3. Since the disorder is autosomal recessive, the parents are carriers of the recessive allele (r) but do not express the disorder themselves because they are homozygous recessive (rr).

4. Therefore, the probability that each of their children will be affected (homozygous recessive, rr) is 1 (or 100%), because they will only inherit the recessive allele from both parents. So, the correct answer is (F) 1.```

One agent solution: ```To solve this, let's consider the possible genotypes and phenotypes of the parents and their offspring. Since both parents are affected by an autosomal recessive disorder, we can represent their genotype as 'rr' (assuming the disorder is caused by a recessive allele 'r' and the dominant allele is 'R').

2. The possible genotypes of their offspring are: - Rr (carrier) - Rr (carrier) 3. Since both parents are affected, the possible genotypes of the parents can be rr.

4. The probability of each child being affected (rr) is the probability of inheriting two recessive alleles, which is 1/4, because each parent has a 1/2 chance of contributing a recessive allele, and the probability of both contributing a recessive allele is (1/2) \* (1/2) = 1/4. So, the probability that each of their children will be affected equals 1/4 or 25%. Therefore, the answer is (I).```

Legend: The red and green shaded text is an example pair for the w/o Semantic Preservation and w/ Semantic Preservation comparison.

Figure 6: Example comparison between w/o semantic preservation (red, brute-force token selection) and w/ semantic preservation (green, semantically coherent expansion)

Preprint. Under review.

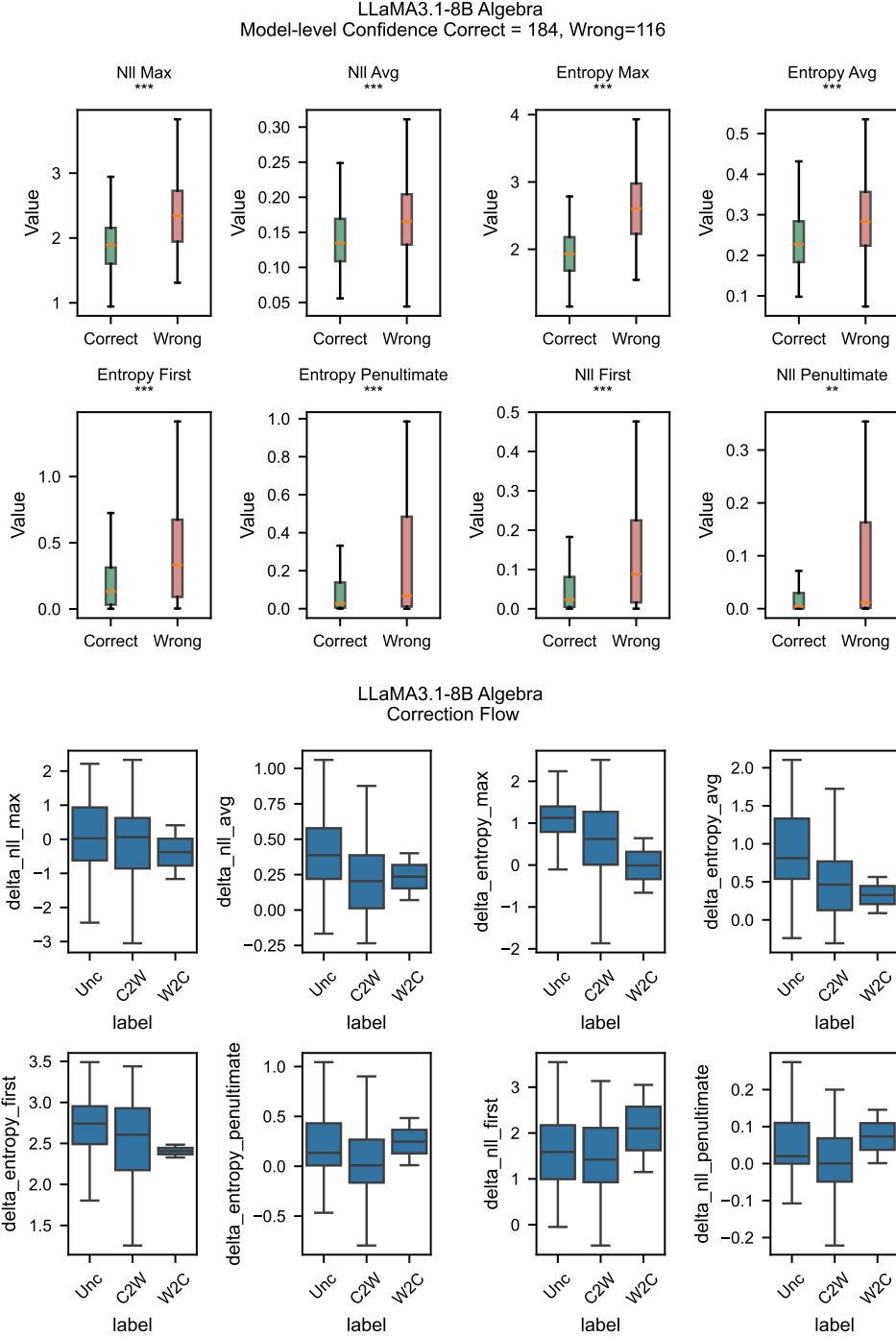


Figure 7: Top: Model-level Confidence result on the Math Algebra dataset with LLaMA3.1-8B. Bottom: Correction flow with 8 deltas of confidence metrics. Unc: the answer remains unchanged between the first round and the last round, C2W: correct to wrong, W2C: wrong to correct.

Preprint. Under review.

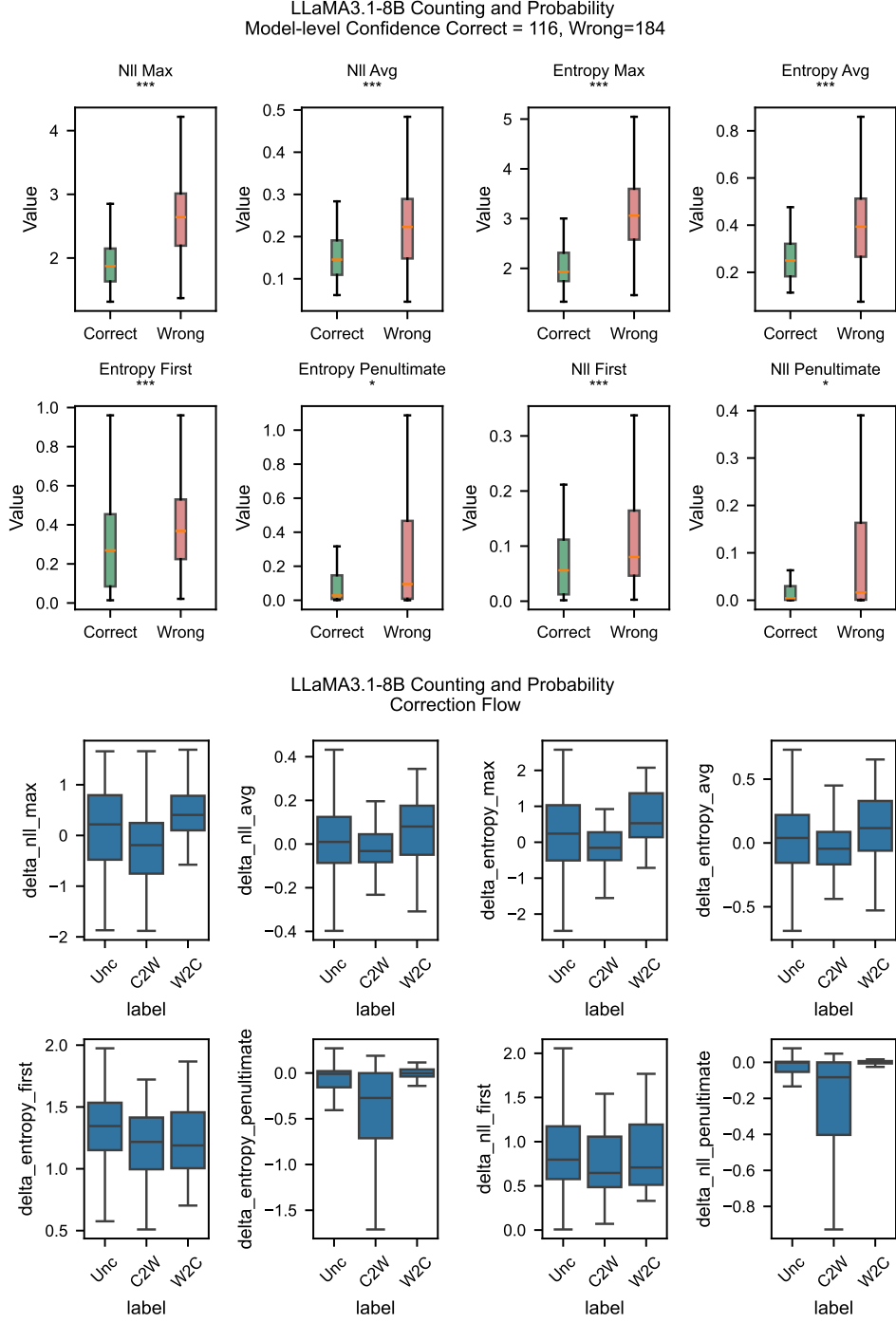


Figure 8: Top: Model-level Confidence result on the Math Counting and Probability dataset with LLaMA3.1-8B. Bottom: Correction flow with 8 deltas of confidence metrics. Unc: the answer remains unchanged between the first round and the last round, C2W: correct to wrong, W2C: wrong to correct.



Preprint. Under review.

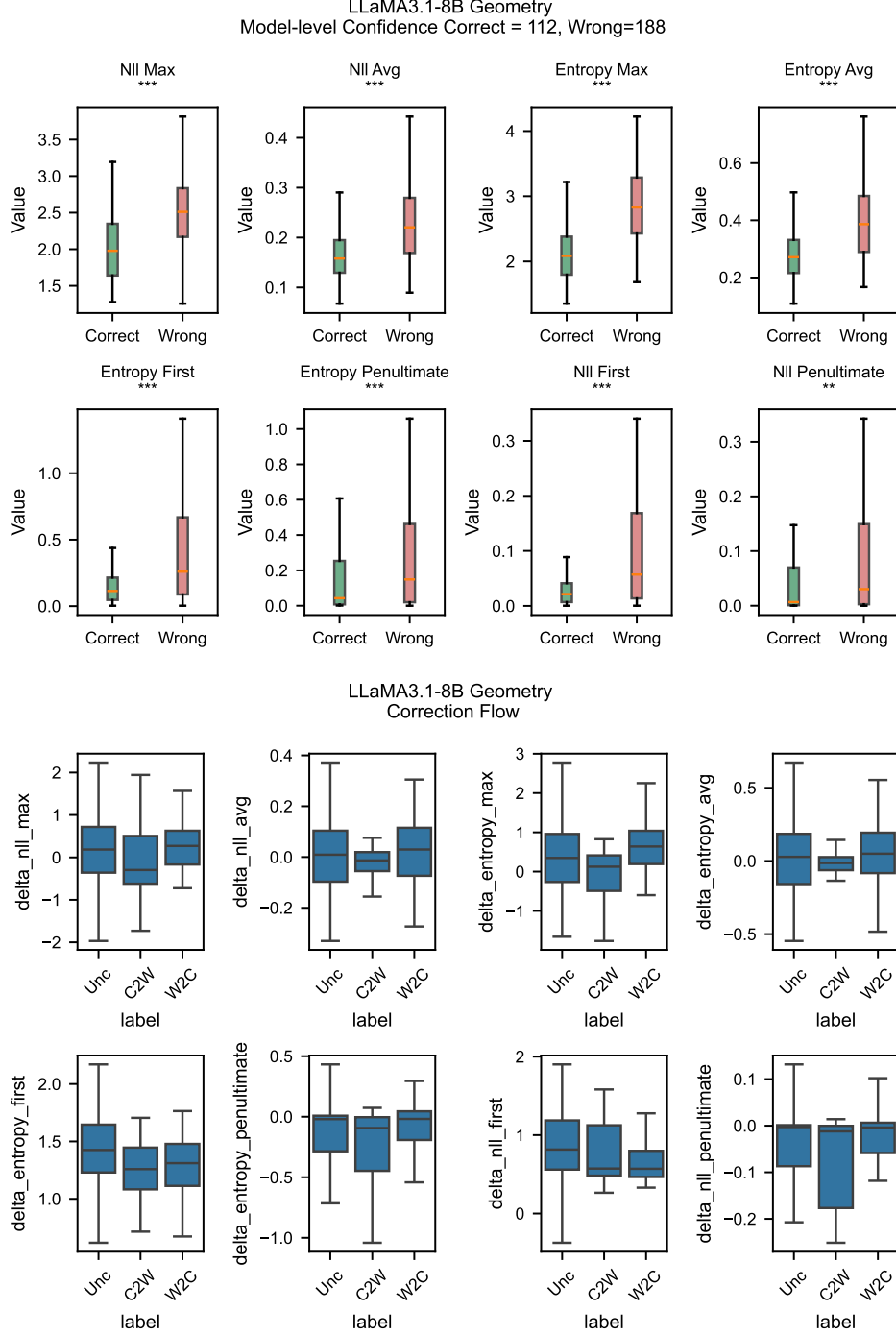


Figure 9: Top: Model-level Confidence result on the Math Geometry dataset with LLaMA3.1-8B. Bottom: Correction flow with 8 deltas of confidence metrics. Unc: the answer remains unchanged between the first round and the last round, C2W: correct to wrong, W2C: wrong to correct.

Preprint. Under review.

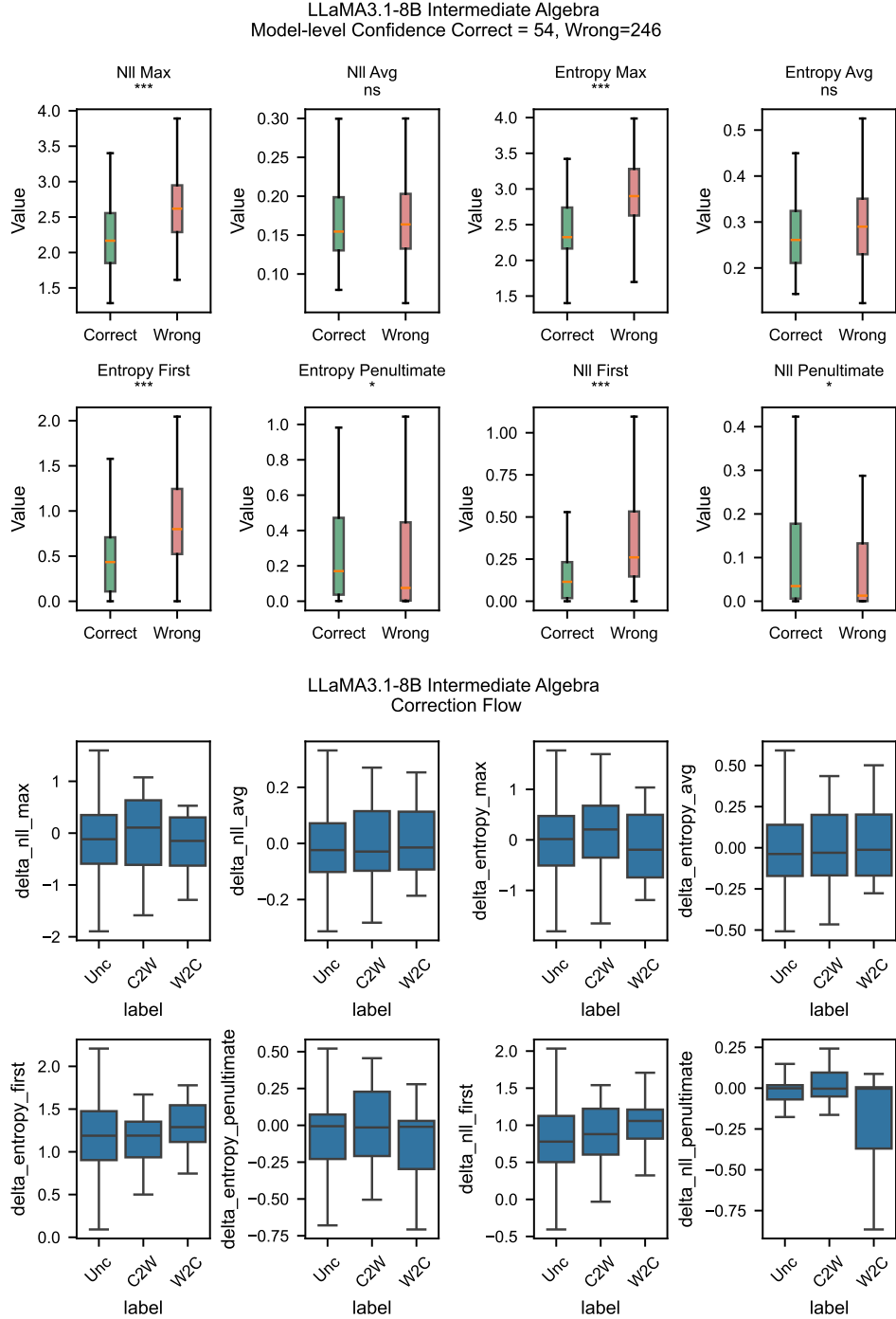


Figure 10: Top: Model-level Confidence result on the Math Intermediate Algebra dataset with LLaMA3.1-8B. Bottom: Correction flow with 8 deltas of confidence metrics. Unc: the answer remains unchanged between the first round and the last round, C2W: correct to wrong, W2C: wrong to correct.

Preprint. Under review.

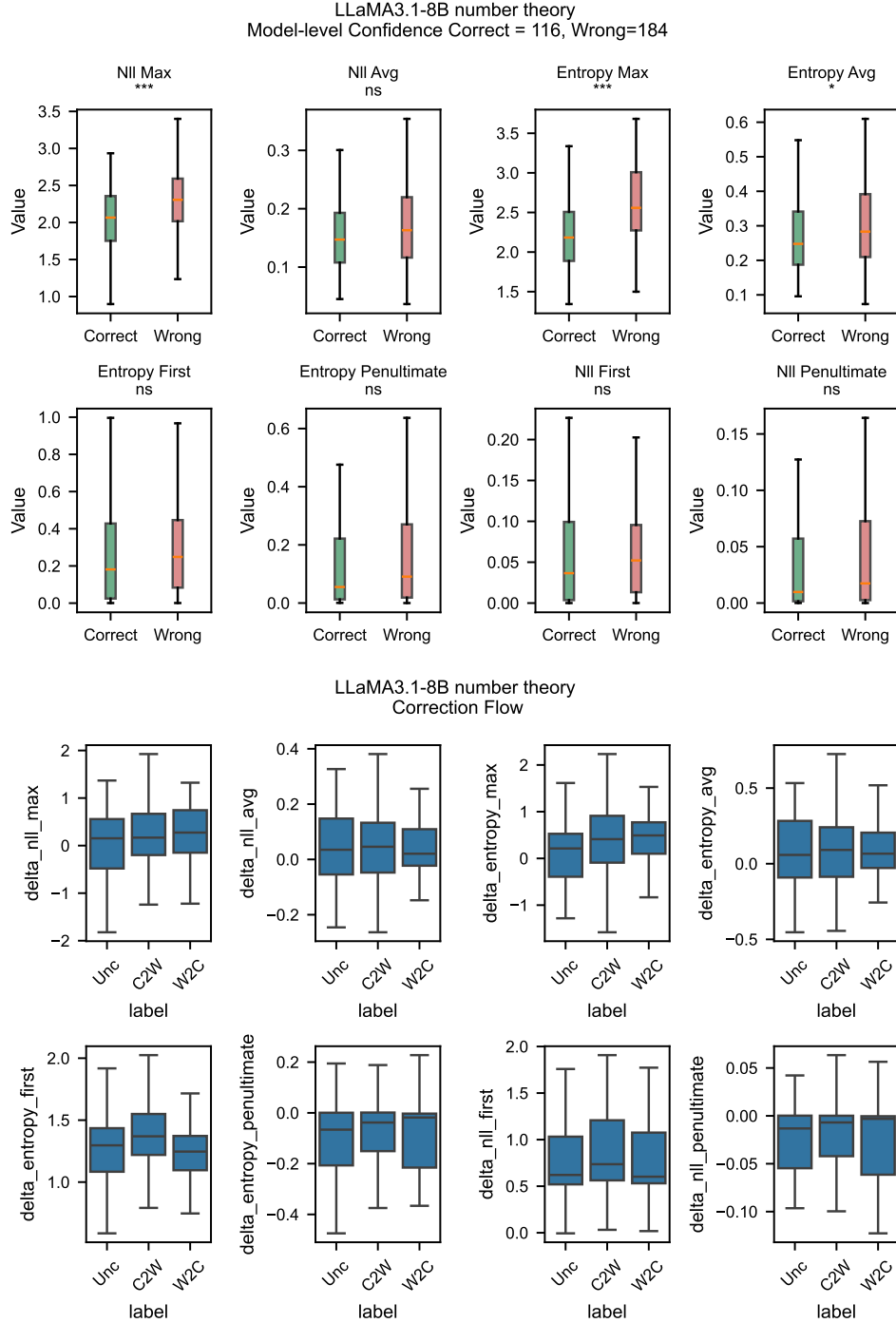


Figure 11: Top: Model-level Confidence result on the Math Number Theory dataset with LLaMA3.1-8B. Bottom: Correction flow with 8 deltas of confidence metrics. Unc: the answer remains unchanged between the first round and the last round, C2W: correct to wrong, W2C: wrong to correct.

Preprint. Under review.

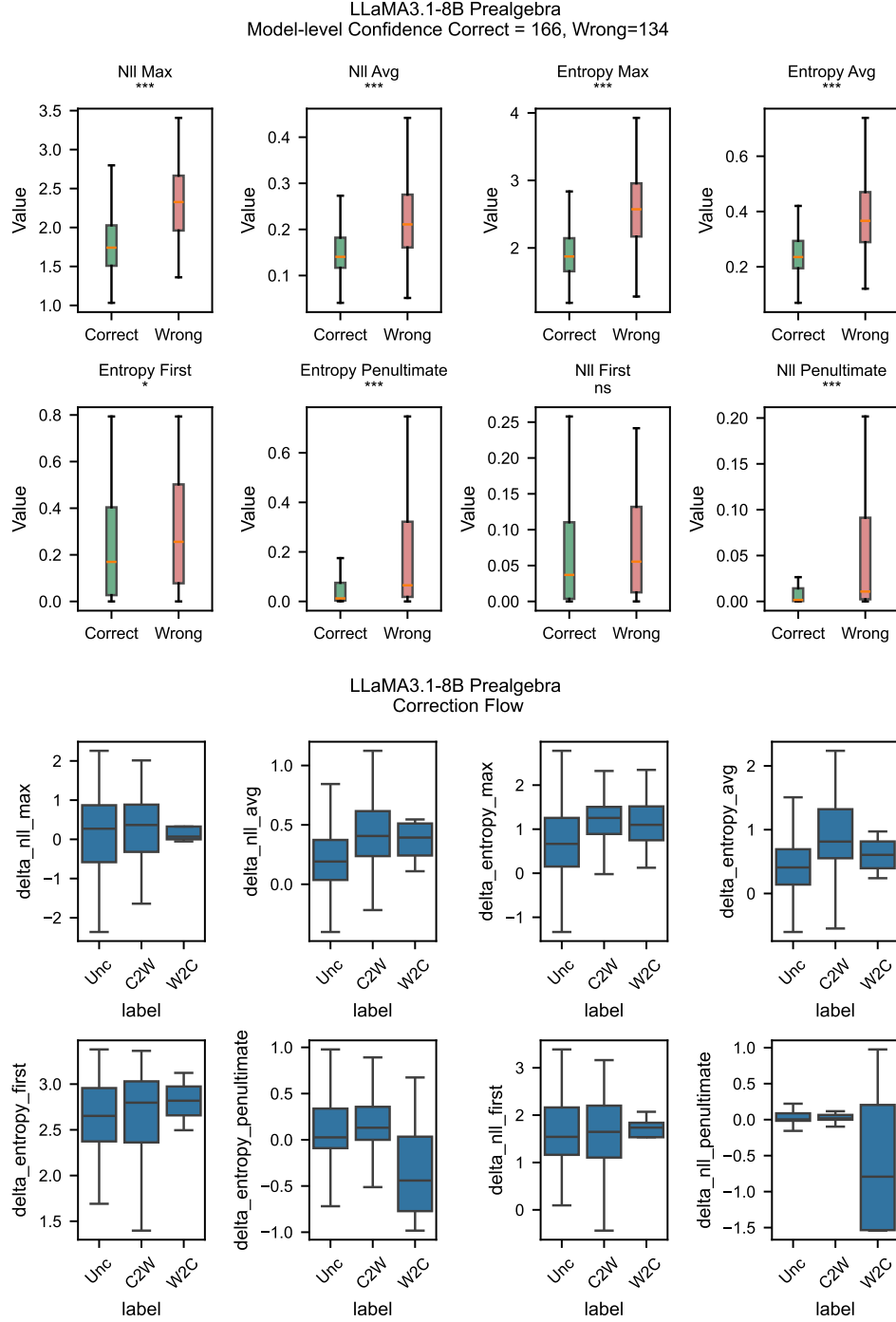


Figure 12: Top: Model-level Confidence result on the Math Prealgebra dataset with LLaMA3.1-8B. Bottom: Correction flow with 8 deltas of confidence metrics. Unc: the answer remains unchanged between the first round and the last round, C2W: correct to wrong, W2C: wrong to correct.

Preprint. Under review.

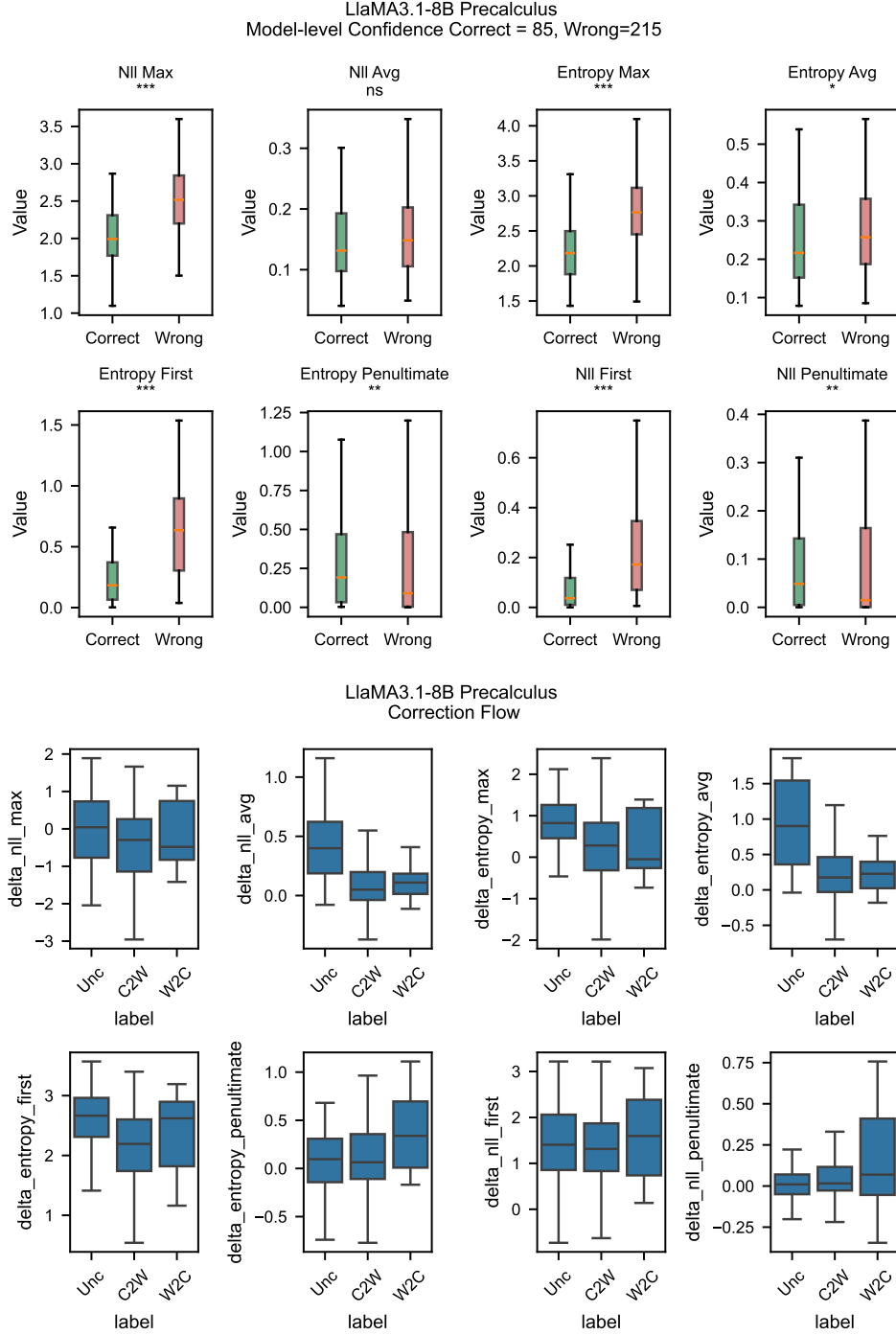


Figure 13: Top: Model-level Confidence result on the Math Precalculus dataset with LLaMA3.1-8B. Bottom: Correction flow with 8 deltas of confidence metrics. Unc: the answer remains unchanged between the first round and the last round, C2W: correct to wrong, W2C: wrong to correct.

Preprint. Under review.

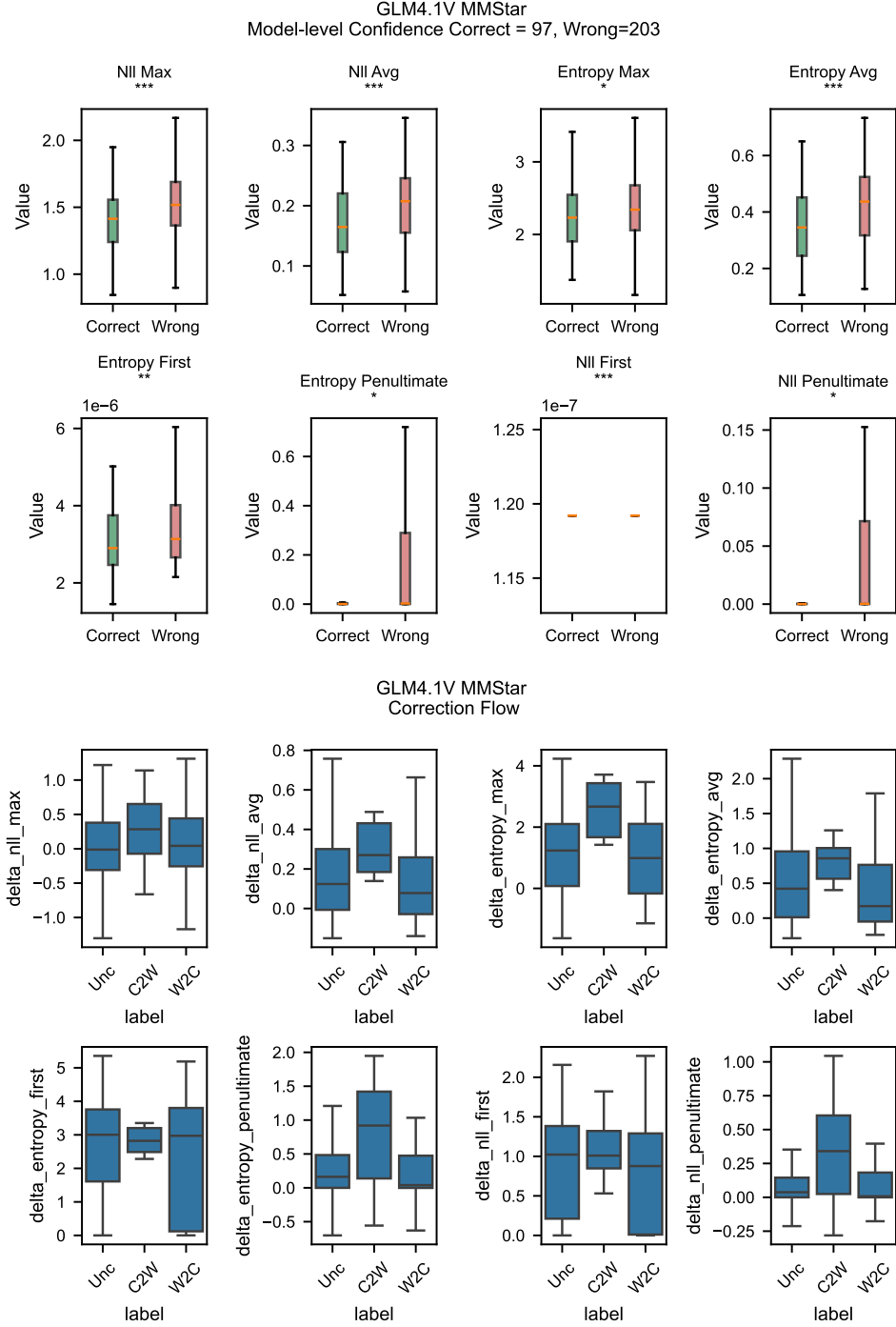


Figure 14: Top: Model-level Confidence result on the MMStar dataset with GLM4.1V. Bottom: Correction flow with 8 deltas of confidence metrics. Unc: the answer remains unchanged between the first round and the last round, C2W: correct to wrong, W2C: wrong to correct.

Preprint. Under review.

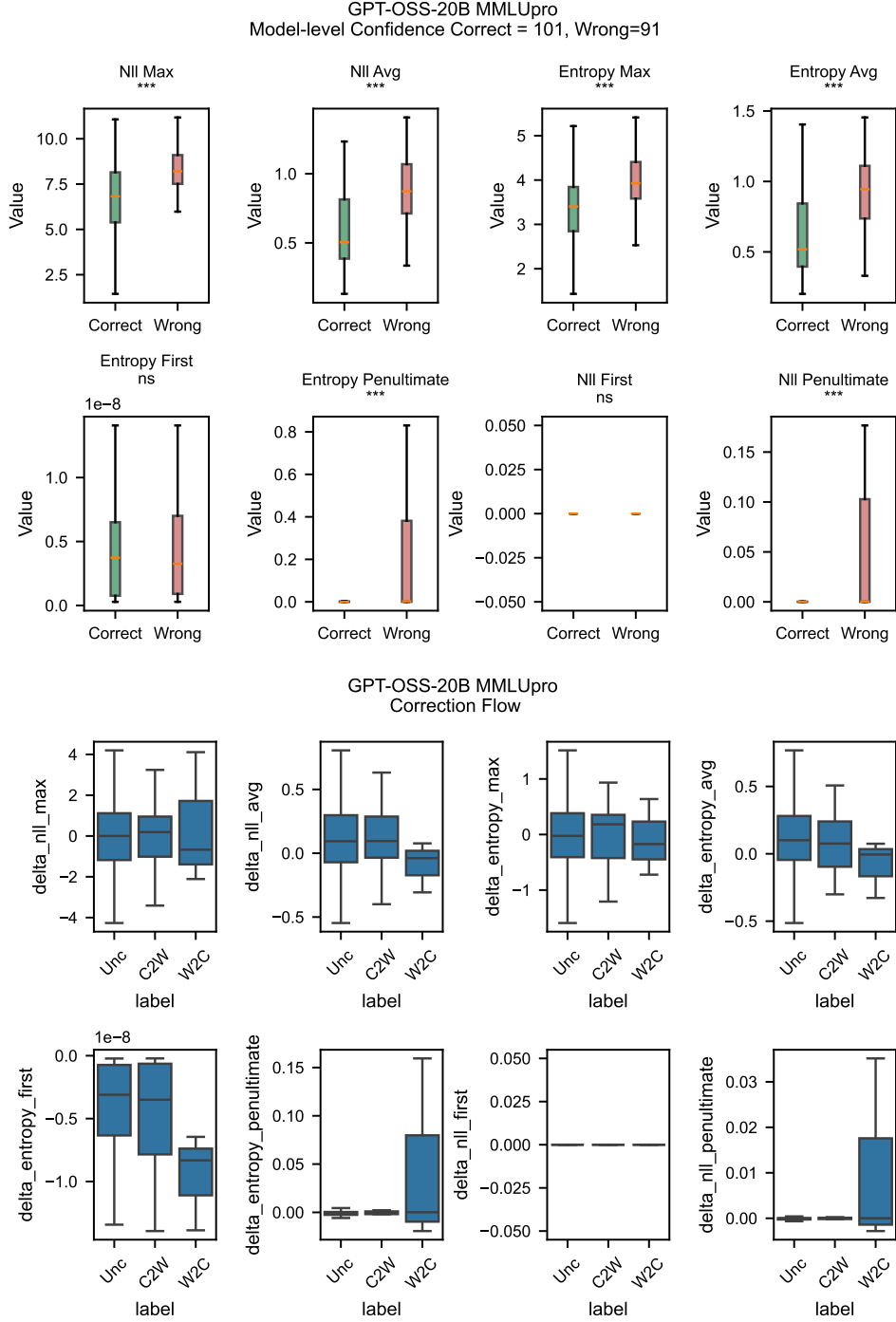


Figure 15: Top: Model-level Confidence result on the MMLUpro dataset with GPT-OSS-20B. Bottom: Correction flow with 8 deltas of confidence metrics. Unc: the answer remains unchanged between the first round and the last round, C2W: correct to wrong, W2C: wrong to correct.

Preprint. Under review.

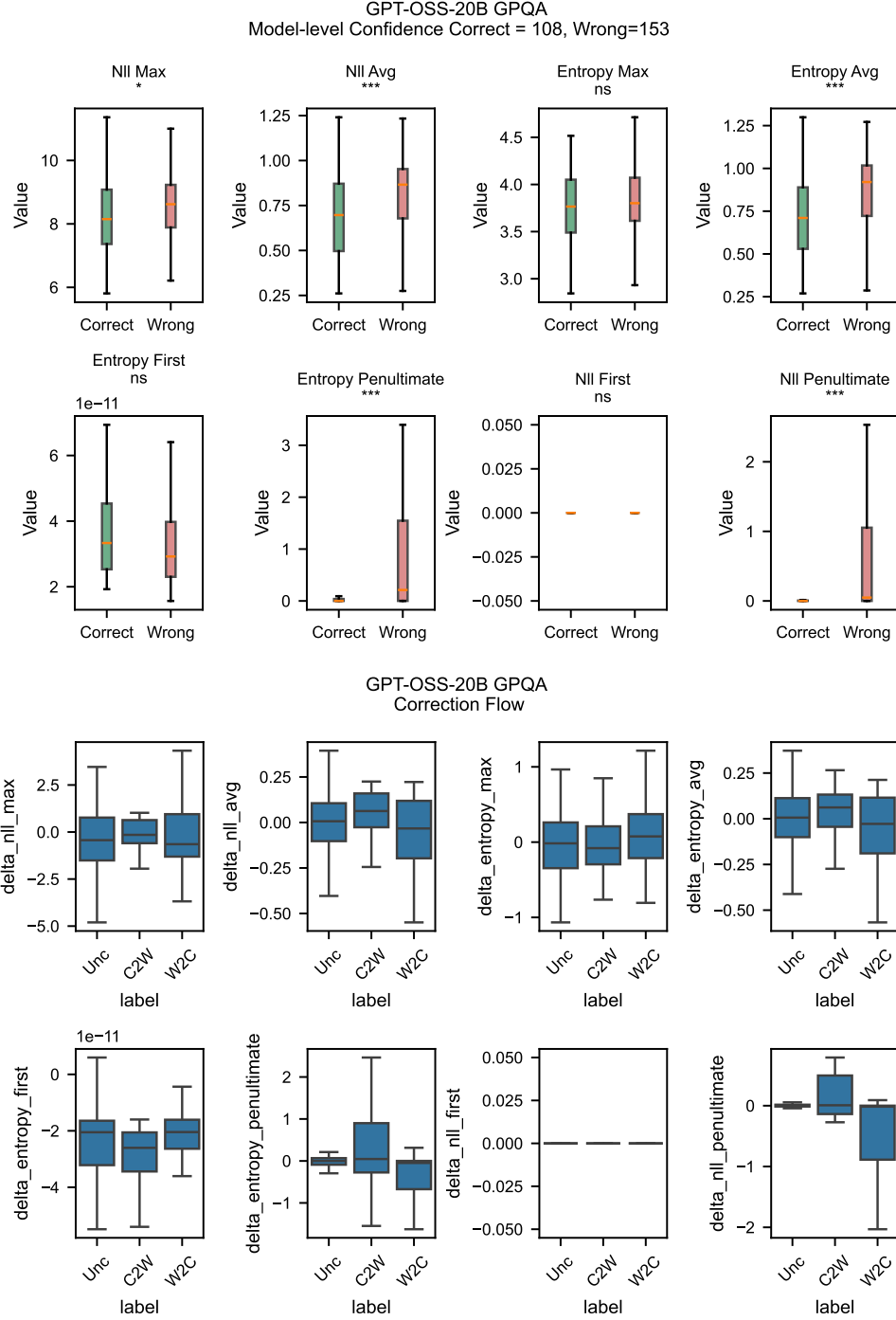


Figure 16: Top: Model-level Confidence result on the GPQA dataset with GPT-OSS-20B. Bottom: Correction flow with 8 deltas of confidence metrics. Unc: the answer remains unchanged between the first round and the last round, C2W: correct to wrong, W2C: wrong to correct.



Preprint. Under review.

### Examples of Model-Level Confidence Exit on MMLUpro

<Question, Correct Answer: B>

Can you answer the following question? If a freight train travels at a speed of 20 miles per hour for 6 hours, how far will it travel?: (A) 90 miles, (B) 120 miles, (C) 150 miles, (D) 140 miles, (E) 26 miles, (F) 100 miles, (G) 80 miles, (H) 110 miles, (I) 160 miles, (J) 12 miles.

<Agent Answer: B> 

To find the distance traveled by the freight train, we need to multiply its speed by the time it traveled.\n\nSpeed = 20 miles per hour\nTime = 6 hours\n\nDistance = Speed x Time\n= 20 miles/hour x 6 hours\n= 120 miles\n\nSo, the freight train will travel 120 miles.\n\nThe correct answer is (B).

<Question, Correct Answer: E>

Can you answer the following question? A driver and passenger were driving to work in the former's automobile one morning. As they were traveling at a speed of 20 m. p. h. (which was within the posted speed limit), the passenger suddenly pointed to an overturned vehicle along the side of the highway and said, "Look at that car upside down. " The driver turned to look at the overturned vehicle. As he was looking toward the side of the road, the driver failed to see an abandoned vehicle with a flat tire in the highway about 200 feet in front of his approaching auto. Seconds later, the driver crashed into the rear of the abandoned auto and was injured. The jurisdiction has a relevant comparative negligence statute in effect. If the driver asserts a claim against the owner of the abandoned auto, the most likely result is that the driver will: (A) recover nothing, because the passenger's act was a supervening cause., (B) recover only a portion of damages, because both the driver and the owner of the abandoned auto were negligent., (C) recover nothing, because he was distracted while driving., (D) recover all of his damages, because the owner of the abandoned auto was negligent., (E) recover only a portion of damages, because the abandoned auto was in plain view., (F) recover all of his damages, because the abandoned auto was a traffic hazard., (G) recover all of his damages, because the defendant created a dangerous condition., (H) recover nothing, because he was driving at a speed that didn't allow him to stop in time., (I) recover nothing, because he had the last clear chance to avoid the collision..

<Agent Answer: B>  [But Judged by Gemini and GPT, both B and E are correct.]

(B) To determine the most likely result, let's break down the situation step by step:\n\n1. The driver was driving at a speed of 20 m.p.h. within the posted speed limit.\n2. The passenger pointed out the overturned vehicle, distracting the driver and causing him to take his eyes off the road.\n3. As a result, the driver failed to see the abandoned vehicle with a flat tire in front of him.\n4. The driver crashed into the abandoned vehicle.\n\nGiven the comparative negligence statute, the jurisdiction likely follows a modified comparative negligence system. In this system, a plaintiff can recover damages even if they were partially at fault, but their recovery is reduced by the percentage of their own negligence.\n\nThe driver's distraction caused by the passenger's comment likely contributed to the accident, making him partially responsible for the collision. However, the owner of the abandoned auto also created a hazard by leaving the vehicle on the road. Therefore, both parties contributed to the accident.\n\nThe most likely result is that the driver will recover only a portion of damages, as both the driver and the owner of the abandoned auto were negligent.

Figure 17: Examples of model-level early exit cases on MMLUpro datasets. The exit questions are correctly answered in principle.

Preprint. Under review.

### Examples of Model-Level Confidence Exit on ScienceQA

**<Question, Correct Answer: C>**

Question:

What is the capital of Kansas?

Options:

(A) Des Moines

(B) Wichita

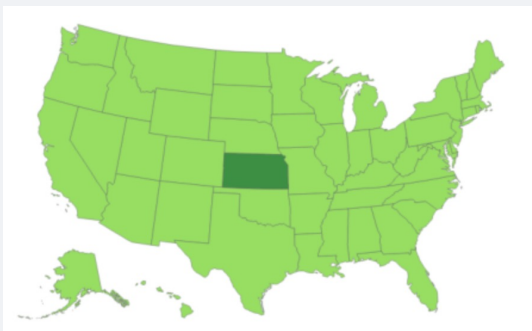
(C) Topeka

(D) Lincoln

Lecture:

**<Agent Answer: C>** ✓

The answer is (C) Topeka.



**<Question, Correct Answer: C>**

Question:

Which country is highlighted?

Options:

(A) Barbados

(B) the Dominican Republic

(C) Saint Lucia

(D) Saint Vincent and the Grenadines

Lecture:

**<Agent Answer: C>** ✓

The answer is (C) Saint Lucia

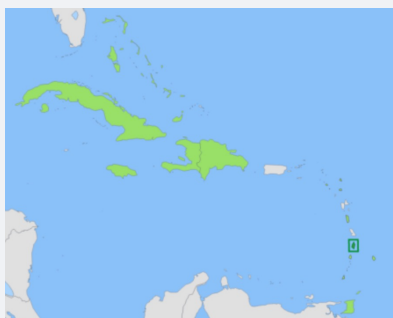


Figure 18: Examples of model-level early exit cases on ScienceQA datasets. The exit questions are correctly answered in principle.

### Examples of Model-Level Confidence Exit on Math

<Question, Correct Answer: [1, 4]>

Question:

Given a mathematics problem, determine the answer.

Question: Compute the domain of the real-valued function  $f(x) = \sqrt{1 - \sqrt{2 - \sqrt{x}}}$ .

Your final answer should be in the form  $\boxed{\text{answer}}$ , at the end of your response.

<Agent Answer: [1, 4]> ✓

To find the domain of the function  $f(x) = \sqrt{1 - \sqrt{2 - \sqrt{x}}}$ , we need to ensure that the expression under the square roots is non-negative. First, we start with the innermost expression,  $\sqrt{x}$ . Since the square root of a number is non-negative, we know that  $x \geq 0$ . Next, we consider the expression  $2 - \sqrt{x}$ . For this to be non-negative, we need  $2 - \sqrt{x} \geq 0$ . Solving for  $\sqrt{x}$ , we get  $\sqrt{x} \leq 2$ , which implies  $x \leq 4$ . Finally, we consider the expression  $1 - \sqrt{2 - \sqrt{x}}$ . For this to be non-negative, we need  $1 - \sqrt{2 - \sqrt{x}} \geq 0$ . Solving for  $\sqrt{2 - \sqrt{x}}$ , we get  $\sqrt{2 - \sqrt{x}} \leq 1$ , which implies  $2 - \sqrt{x} \leq 1$ . Simplifying, we get  $\sqrt{x} \geq 1$ , which implies  $x \geq 1$ . Combining the conditions, we have  $1 \leq x \leq 4$ . Therefore, the domain of the function is  $\boxed{[1, 4]}$ .

<Question, Correct Answer: 3>

Given a mathematics problem, determine the answer.

Question: Evaluate  $\left\lceil \left( \frac{3}{2} \right)^2 + \left\lceil \left( \frac{3}{2} \right)^2 \right\rceil \right\rceil$ .

Your final answer should be in the form  $\boxed{\text{answer}}$ , at the end of your response.

<Agent Answer: 9> ✗

To evaluate this expression, we need to follow the order of operations (PEMDAS):

- Evaluate the expressions inside the parentheses:  $\left( \frac{3}{2} \right)^2 = \frac{9}{4} = 2.25$ .
- Evaluate the ceiling function of the result:  $\left\lceil 2.25 \right\rceil = 3$ .
- Evaluate the expression inside the other set of parentheses:  $\frac{3}{2}$ .
- Evaluate the ceiling function of the result:  $\left\lceil \frac{3}{2} \right\rceil = 2$ .
- Square the result of step 3:  $\left( \frac{3}{2} \right)^2 = \frac{9}{4} = 2.25$ .
- Add the results of steps 2 and 5:  $3 + 2.25 = 5.25$ .
- Evaluate the ceiling function of the result of step 6:  $\left\lceil 5.25 \right\rceil = 6$ .
- Now, add the results of steps 2 and 7:  $3 + 6 = 9$ .

Therefore, the final answer is  $\boxed{9}$ .

Figure 19: Examples of model-level early exit cases on Math datasets. The bottom overconfident case shows the model partially analyzes the question, potentially leading to a high confidence.

### Examples of Model-Level Confidence Exit on GPQA

#### <Question, Correct Answer: B>

Can you answer the following question? In the vacuum, we have the following mixture

$$\left| \nu_{\text{left}} \right\rangle = e^{i p_1 x} \cos \theta \left| \nu_1 \right\rangle + e^{i p_2 x} \sin \theta \left| \nu_2 \right\rangle$$

where  $\theta$  is the mixing angle, and  $\left| \nu_1 \right\rangle$  and  $\left| \nu_2 \right\rangle$  are the basis of mass eigenstates.

At what value of the mixing angle we will obtain the transition probability  $P(\nu_e \rightarrow \nu_{\mu}) = 1$ .

Choices: (A)  $\pi/2$  (B)  $\pi/4$  (C)  $\pi$  (D)  $\pi/3$ .

#### <Agent Answer: B> ✓

analysis: We need to analyze neutrino flavor mixing: initial state is electron neutrino? They give general state  $|\nu(x)\rangle = e^{i p_1 x} \cos \theta |\nu_1\rangle + e^{i p_2 x} \sin \theta |\nu_2\rangle$ . The probability of transition from  $\nu_e$  to  $\nu_{\mu} = \sin^2 2\theta \sin^2(\Delta m^2 L / (4E))$ . For maximum probability 1, we need  $\sin^2 2\theta = 1$  AND  $\sin^2$  phase term = 1 (i.e., argument =  $\pi/2 + n\pi$ ). But question asks at what mixing angle we will obtain the probability 1 independent of oscillation? If you want probability 1, you need perfect mixing:  $\theta = \pi/4$  gives maximal mixing so  $\sin^2 2\theta = \sin^2(\pi/2) = 1$ . So answer (B)  $\pi/4$ . So "The correct answer is (B)".

assistant: In a two-flavor framework the transition probability is  $P(\nu_e \rightarrow \nu_{\mu}) = \sin^2 2\theta \sin^2(\frac{\Delta m^2 L}{4E})$  where  $\theta$  is the mixing angle and the second sine factor contains the usual oscillation phase. To make the probability equal to unity for some distance  $L$  we need both factors to be unity. The phase factor can become unity for suitable  $L$  (e.g. when  $\frac{\Delta m^2 L}{4E} = \pi/2$ ). The remaining requirement is therefore  $\sin^2 2\theta = 1$ .  $\Rightarrow \sin 2\theta = 1$ .  $\Rightarrow 2\theta = \pi/2$ .  $\Rightarrow \theta = \pi/4$ . Among the provided options, the value that gives complete mixing is  $\theta = \pi/4$ . Therefore the correct choice is (B). The correct answer is (B).

#### <Question, Correct Answer: B>

Can you answer the following question? In a specific region of the sky, astronomers have observed that the number of stars varies with parallax as  $1/p^5$ . How does the number of stars in that region of the sky change with distance (per unit range of distance,  $r$ )? Choices: (A)  $\sim r^4$  (B)  $\sim r^3$  (C)  $\sim r^2$  (D)  $\sim r^5$ .

#### <Agent Answer: B> ✓

analysis: We need to understand relationship between parallax and distance: parallax  $p$  (in arcseconds) approx  $1/d$  (pc) maybe? For small parallax:  $p = 1/d$  in arcsec if distance in parsecs. So  $d = 1/p$ . Number of stars varying as  $1/p^5$ . Number of stars per parallax interval? They say number of stars varies with parallax as  $1/p^5$ . That implies count  $N(p) \sim 1/p^5$ . We want how number per unit range of distance  $r$  behaves:  $N(r)$  per unit distance? The distribution of stars per unit distance range:  $dN/dr$ . The relation between  $p$  and  $r$ :  $p = 1/r$  (assuming  $p$  in arcsec,  $r$  in parsecs). So  $r = 1/p$ . So  $p = 1/r$ . So  $N(p) \sim 1/p^5$ . But  $N(p)$  is number in certain parallax interval maybe? We need convert to  $N(r)$ . Suppose number per parallax interval  $dp$  is proportional to  $1/p^5$ . The number per unit  $r$ :  $dN/dr = dN/dp \cdot dp/dr$ .  $dN/dp = k \cdot 1/p^5$ .  $dp/dr = d(1/r)/dr = -1/r^2$ . We take magnitude:  $|dp/dr| = 1/r^2$ . Express  $p$  in terms of  $r$ :  $p = 1/r$ . So  $(1/p^5) = 1/(1/r)^5 = r^5$ . So  $dN/dp = k \cdot r^5$ . Multiply by  $|dp/dr| = 1/r^2$  gives  $dN/dr = k \cdot r^5 \cdot (1/r^2) = k \cdot r^3$ . So number per unit distance  $\sim r^3$ . So answer (B). That matches typical uniform space density:  $N(r) \sim r^3$ . So The correct answer is (B).

assistant: The relationship between parallax  $p$  and distance  $r$  is  $p = 1/r$  (with  $p$  in arcseconds and  $r$  in parsecs). The number of stars as a function of parallax is given as  $N(p) \propto 1/p^5$ . To find the differential number per unit distance we use the chain rule  $\frac{dN}{dr} = \frac{dN}{dp} \cdot \frac{dp}{dr}$ . From  $p = 1/r$ , we have  $\frac{dp}{dr} = -1/r^2$ ; taking the magnitude,  $|\frac{dp}{dr}| = 1/r^2$ . Substitute  $p = 1/r$  into  $\frac{dN}{dp}$ :  $\frac{dN}{dp} \propto \frac{1}{(1/r)^5} = r^5$ . Thus,  $\frac{dN}{dr} \propto r^5 \cdot \frac{1}{r^2} = r^3$ . Hence, the number of stars per unit range of distance  $r$  scales as  $r^3$ . The correct answer is (B). (We did not find negative cases when using model-level early exit.)

Figure 20: Examples of model-level early exit cases on GPQA datasets, where we did not find negative cases (i.e., the model is overconfident).

Preprint. Under review.

## Examples of Token-Level Compression

### <Token-Level Semantic Focus Analysis>

Below are the key points where other agents "disagree" with your own reasoning.  
Concentrate on those disagreements and Keep or revise your answer accordingly.

<Copy the first agent message as the content is determined by internal attention that would be useful for current agent>

**One agent solution:** We need interpret reaction: 9-fluorenone + 2 equivalents of Selectfluor under photochemical? 400nm? Then product observed via <sup>19</sup>F NMR with two fluorine signals: +29.8 ppm and -115.7 ppm. Which product matches? We need consider structure of products.

9-fluorenone has no fluorine. Fluorine signals appear from added fluorine atoms (from Selectfluor) presumably fluorination of the aromatic system. <sup>19</sup>F NMR chemical shifts: +29.8 ppm is typical for aromatic fluorine attached to sp<sup>2</sup> carbon in aromatic ring (like fluorobenzene ~7-8 ppm? wait fluorobenzene is 7 ppm. Actually <sup>19</sup>F NMR for fluoride attached to benzene ring: about -113 to -115 ppm. For CF<sub>3</sub> group ~ -65 ppm. For fluorines bound to carbonyl? Let's recall typical shifts: In 1,2-difluoro compounds, one signal can be upfield and one downfield depending on environment. For example, 2,4-difluoroaniline maybe at ~-110 and -45? Hard.

Let's analyze options:

(A) 1,5-difluoro-9H-fluoren-9-one. This would be fluorine at positions 1 and 5 of the fluorenone core. Aromatic fluorine signals for 1,5-difluoro; symmetrical? Two identical signals? But <sup>19</sup>F NMR would show one signal (if equivalent)? Actually if two fluorines are meta to each other? Might be equivalent? 1,5-difluoro: two positions are equivalent if molecule symmetrical (flipped). For 1,5-difluoro-9-fluorenone, the two fluorines would be in symmetric positions. Would give a single <sup>19</sup>F signal. But question says two fluorine signals, so (A) unlikely.

(C) 2,6-difluoro-9H-fluoren-9-one: also symmetrical? 2,6 are para on each phenyl ring? The two fluorines are in equivalent positions (mirror plane through center). Might again give single signal. So (C) unlikely.

(B) 2'-fluoro-[1,1'-biphenyl]-2-carbonyl fluoride. This is a different product: a biphenyl with fluorine on one ring and a carbonyl fluoride on other. This would have a single fluorine (the one at 2' position). Therefore only one <sup>19</sup>F signal. So also unlikely.

(D) 4'-fluoro-[1,1'-biphenyl]-2-carbonyl fluoride. Also single fluorine.

...

**One agent solution:** We need to analyze the described reaction: 9-fluorenone (structure: a fluorenone: a fused ring (fluorene) with a carbonyl at center? Actually 9-fluorenone is a 9-carbonyl in fluorene. 2 equivalents of Selectfluor used in acetonitrile, irradiated with 400 nm light. Reaction product has two fluorine signals: +29.8 ppm and -115.7 ppm in <sup>19</sup>F NMR. Need to identify which structure among the options is correct product. Let's parse the options:

(A) 1,5-difluoro-9H-fluoren-9-one

(B) 2'-fluoro-[1,1'-biphenyl]-2-carbonyl fluoride

(C) 2,6-difluoro-9H-fluoren-9-one

(D) 4'-fluoro-[1,1'-biphenyl]-2-carbonyl fluoride

<Delete useless background information>

We need to consider the reaction conditions. Selectfluor is used as a fluorination agent. Selectfluor is known to be a strong oxidant and fluorinating reagent, can generate radical cation. 9-Fluorenone has a central carbonyl. But 2 equivalents of Selectfluor might produce two fluorine atoms.

<Keep useful induction>

Looking at the product options: Both (B) and (D) are biphenyl derivatives with a carbonyl fluoride group at the 2-position of the biphenyl core. The difference is the position of a fluorine substituent on the biphenyl ring: 2' vs 4'.

<Cut out useless reasoning paths>

Options (A) and (C) are difluoro-substituted fluorene-9-one. 1,5-difluoro vs 2,6-difluoro arrangement on the fluorene core. Those are fluorinated at the aromatic positions adjacent to the carbonyl (perhaps positions that are 1 and 5, or 2 and 6 referencing fluorene numbering). But do we expect difluorination of the aromatic ring? Maybe the reaction could be electrophilic aromatic substitution with electrophilic fluorine; but Selectfluor is a fluorinating agent, providing F<sup>+</sup>. With 9-fluorenone, electrophilic aromatic substitution would likely occur at positions ortho/para to the carbonyl? etc) perhaps the substituents on the ring positions adjacent to the central carbonyl. I need to draw structure:

The numbering scheme: positions 1, 3, 5, 7 correspond to the outer ring carbons; Let's quickly sketch: The central carbonyl is at position 9. I need to be precise. Let's step aside: We might not need exact numbering...

Using the reasoning from other agents as additional advice, can you give an updated answer? Examine your solution and that other agents step by step. Let's think step by step. Your last sentence should be 'The correct answer is (insert answer here).' e.g., The correct answer is (A).

Figure 21: Detailed example of token-level semantic focus. There is content from two agents in the figure. The first agent's answer was largely adopted, whereas the second agent's answer was largely cropped. ¶ shows places that were adaptively compressed by the model's attention mechanism.

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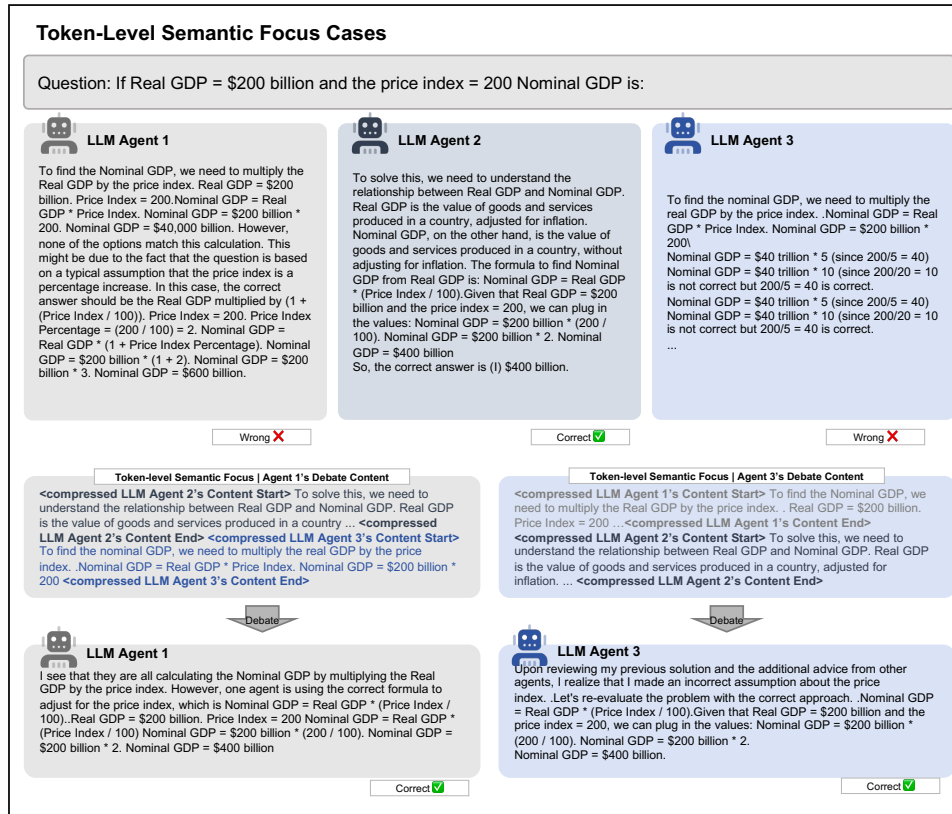


Figure 22: Visualization of token-level semantic focus helped agents correct their answers during the SID. Specifically, SID invokes a compressed debate round, highlighting disagreement-relevant spans across debate contents. Agent 1 and Agent 3 iteratively revise their reasoning based on focused inputs, ultimately correcting earlier errors and converging on the correct answer.